

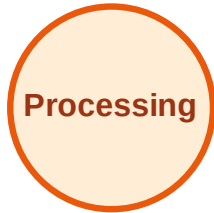
BFS Traversal

Step 1: Start at A (Seeking target node K)

Legend



Complete



Processing



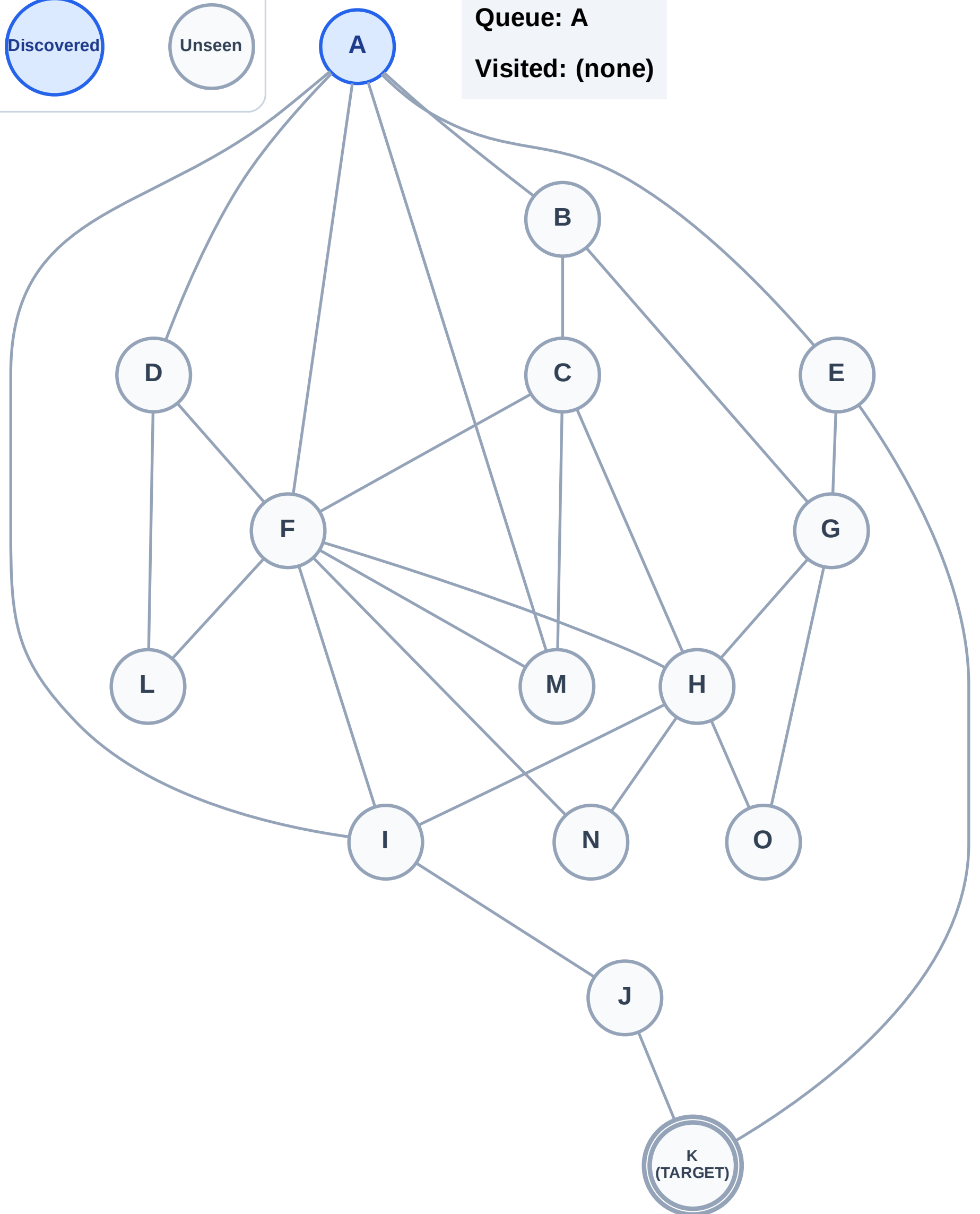
Discovered



Unseen

Queue: A

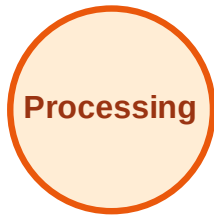
Visited: (none)



BFS Traversal

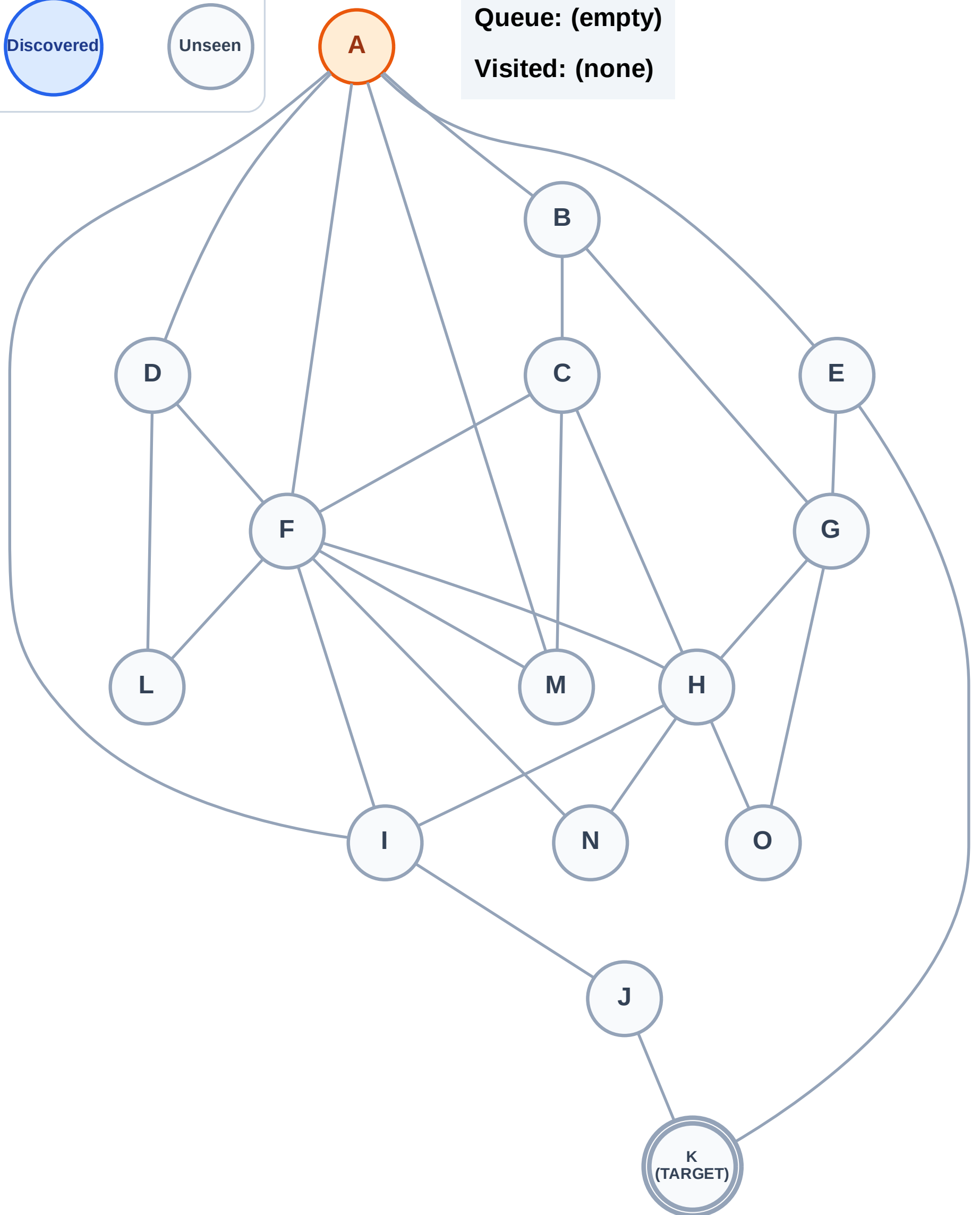
Step 2: Process node A

Legend



Queue: (empty)

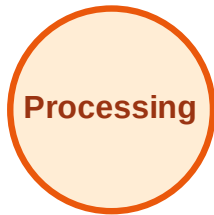
Visited: (none)



BFS Traversal

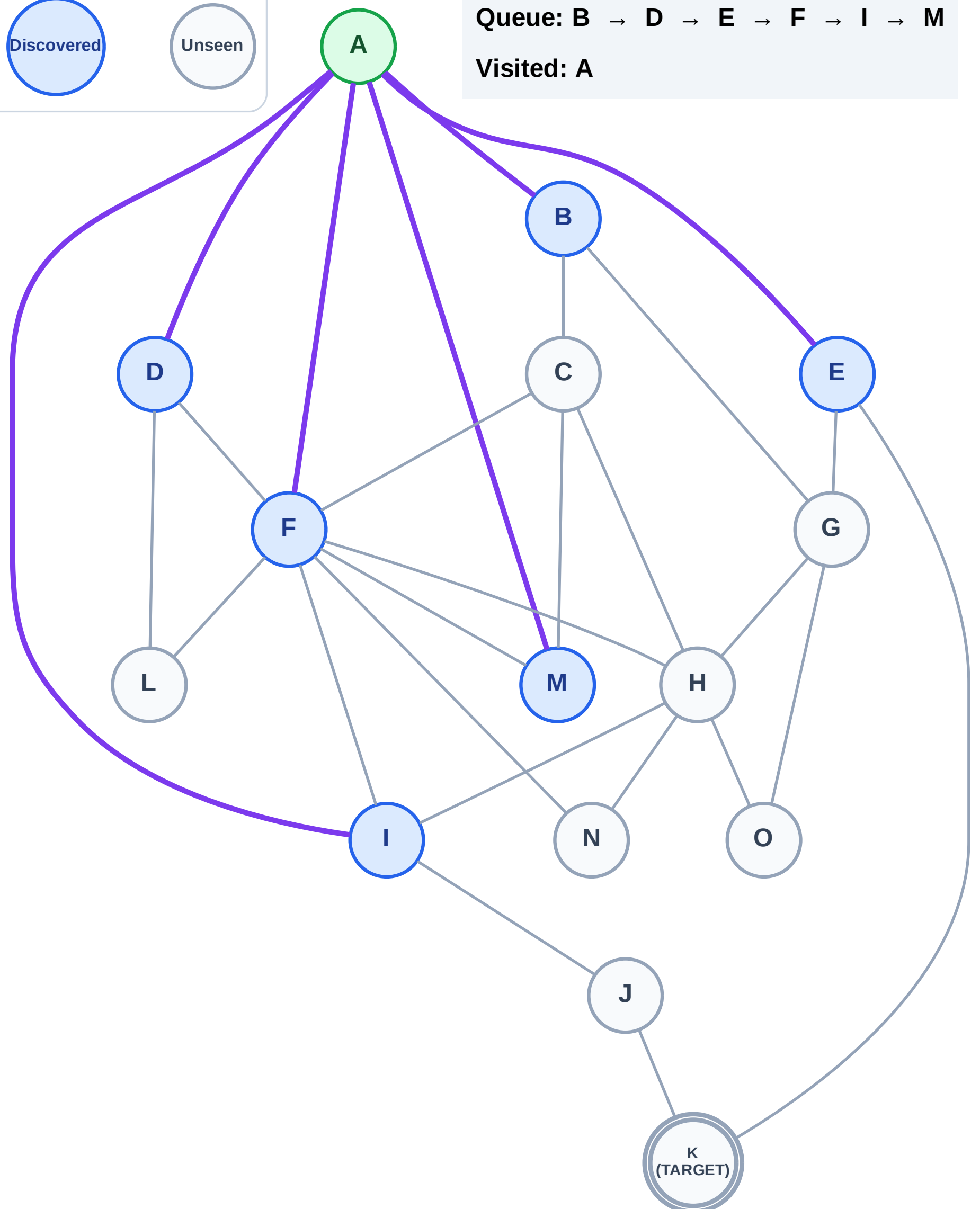
Step 3: Discover B, D, E, F, I, M from A

Legend



Queue: B → D → E → F → I → M

Visited: A



BFS Traversal

Step 4: Process node B

Legend

Complete

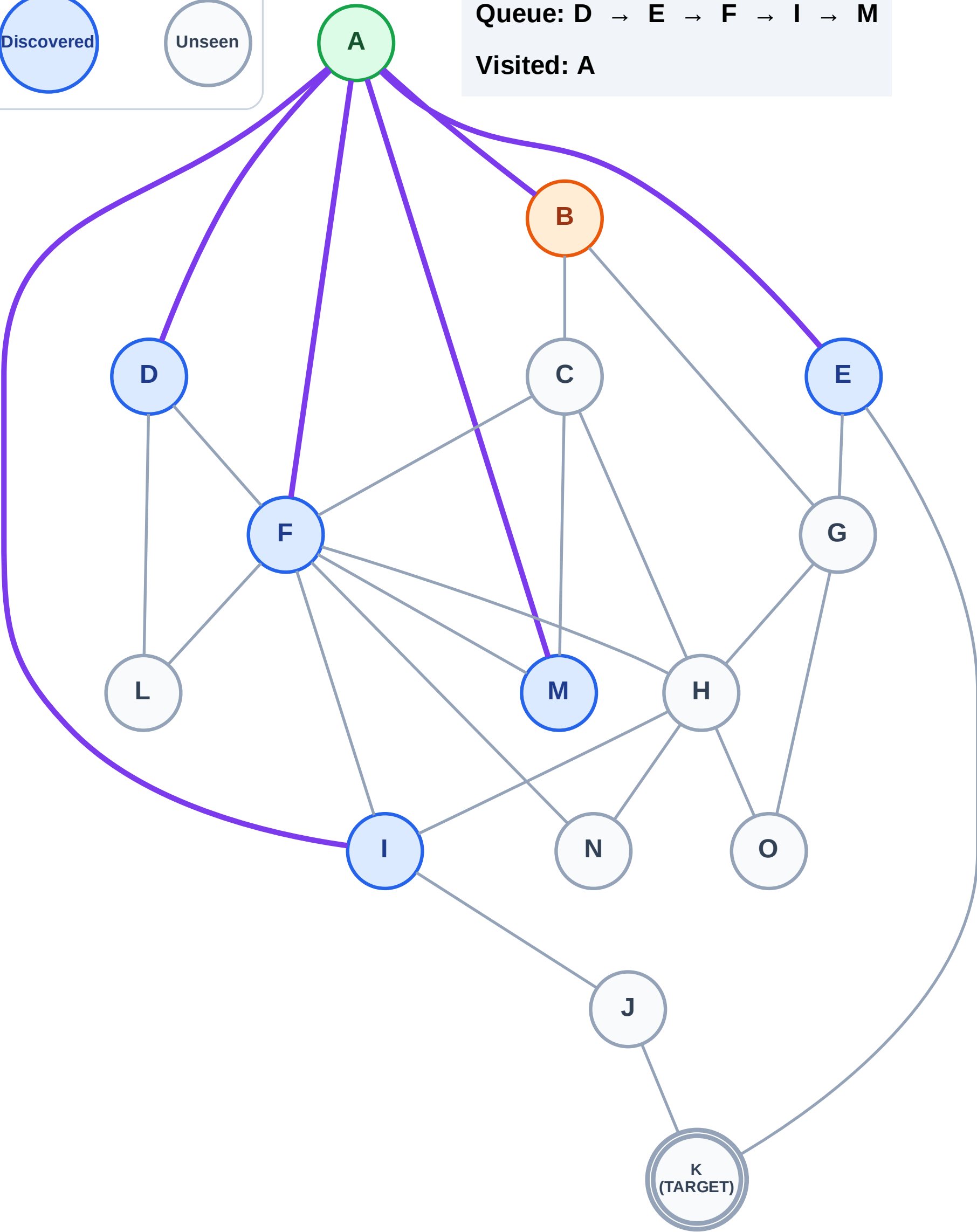
Processing

Discovered

Unseen

Queue: D → E → F → I → M

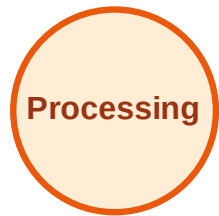
Visited: A



BFS Traversal

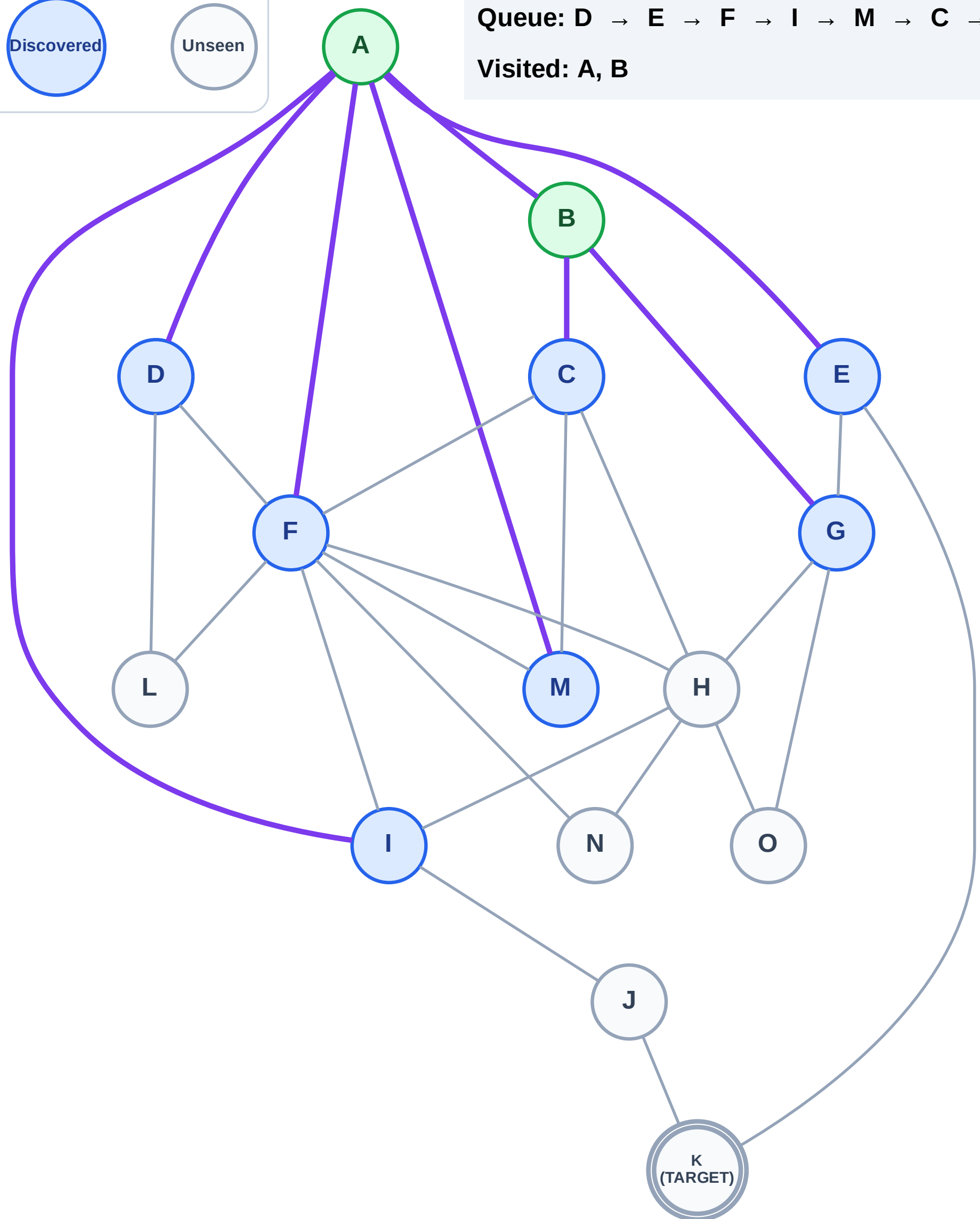
Step 5: Discover C, G from B

Legend



Queue: D → E → F → I → M → C → G

Visited: A, B



BFS Traversal

Step 6: Process node D

Legend

Complete

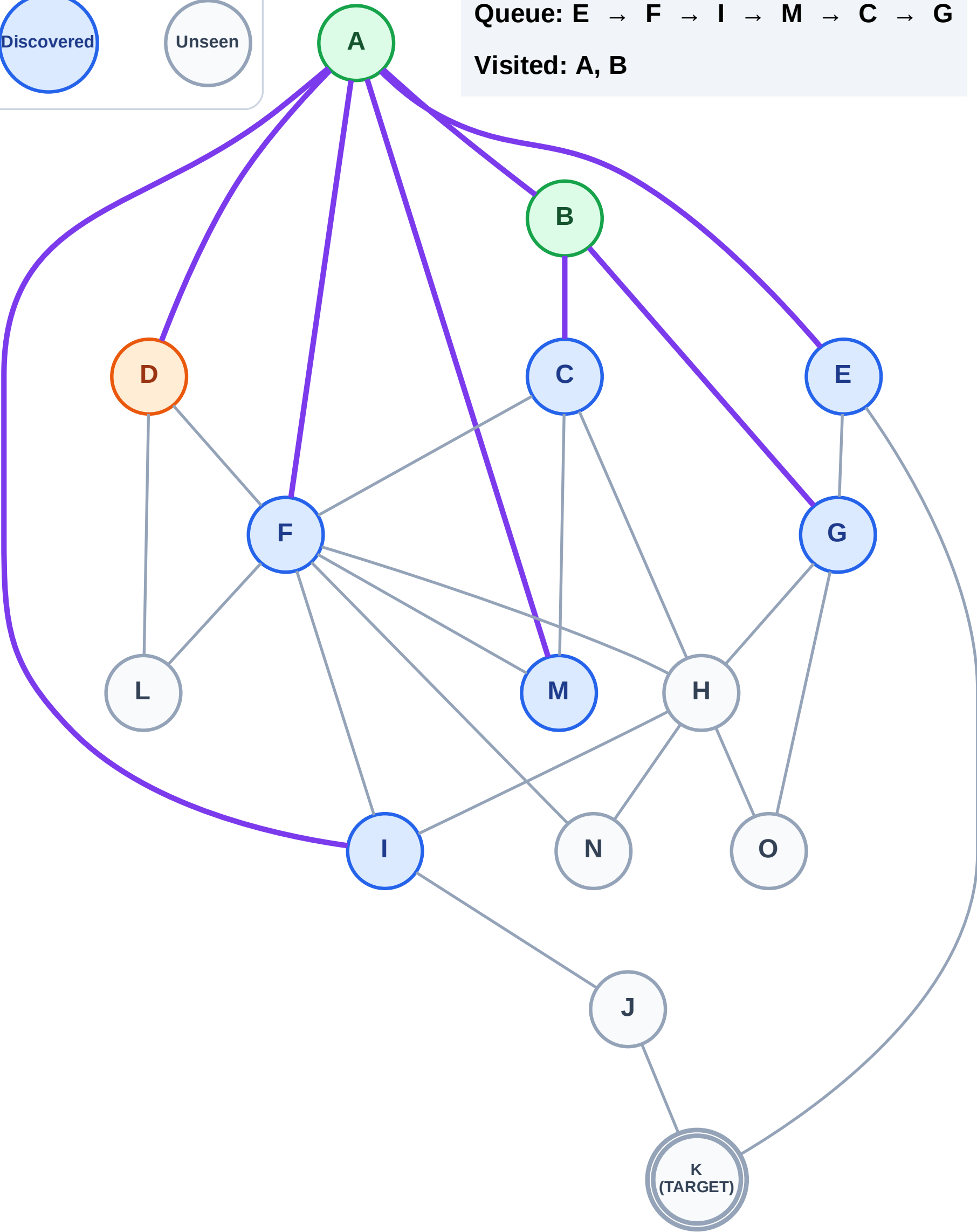
Processing

Discovered

Unseen

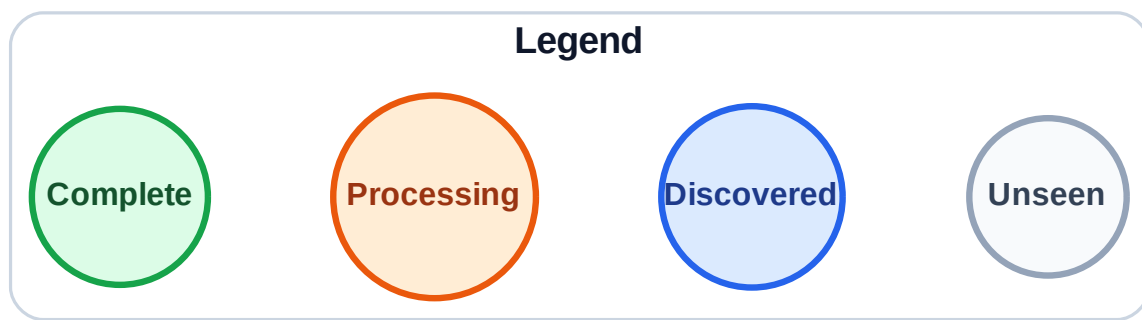
Queue: E → F → I → M → C → G

Visited: A, B



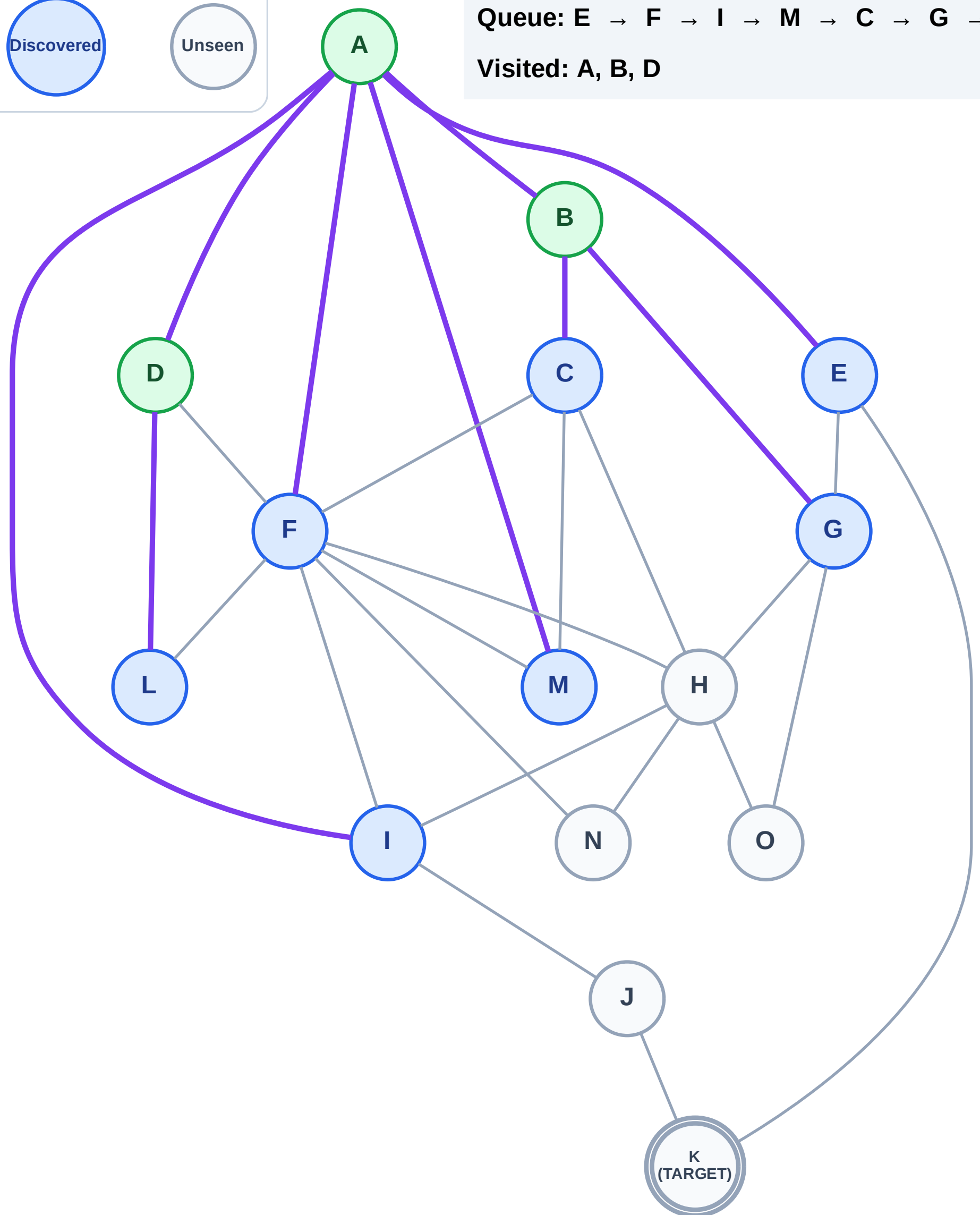
BFS Traversal

Step 7: Discover L from D



Queue: E → F → I → M → C → G → L

Visited: A, B, D



BFS Traversal

Step 8: Process node E

Legend

Complete

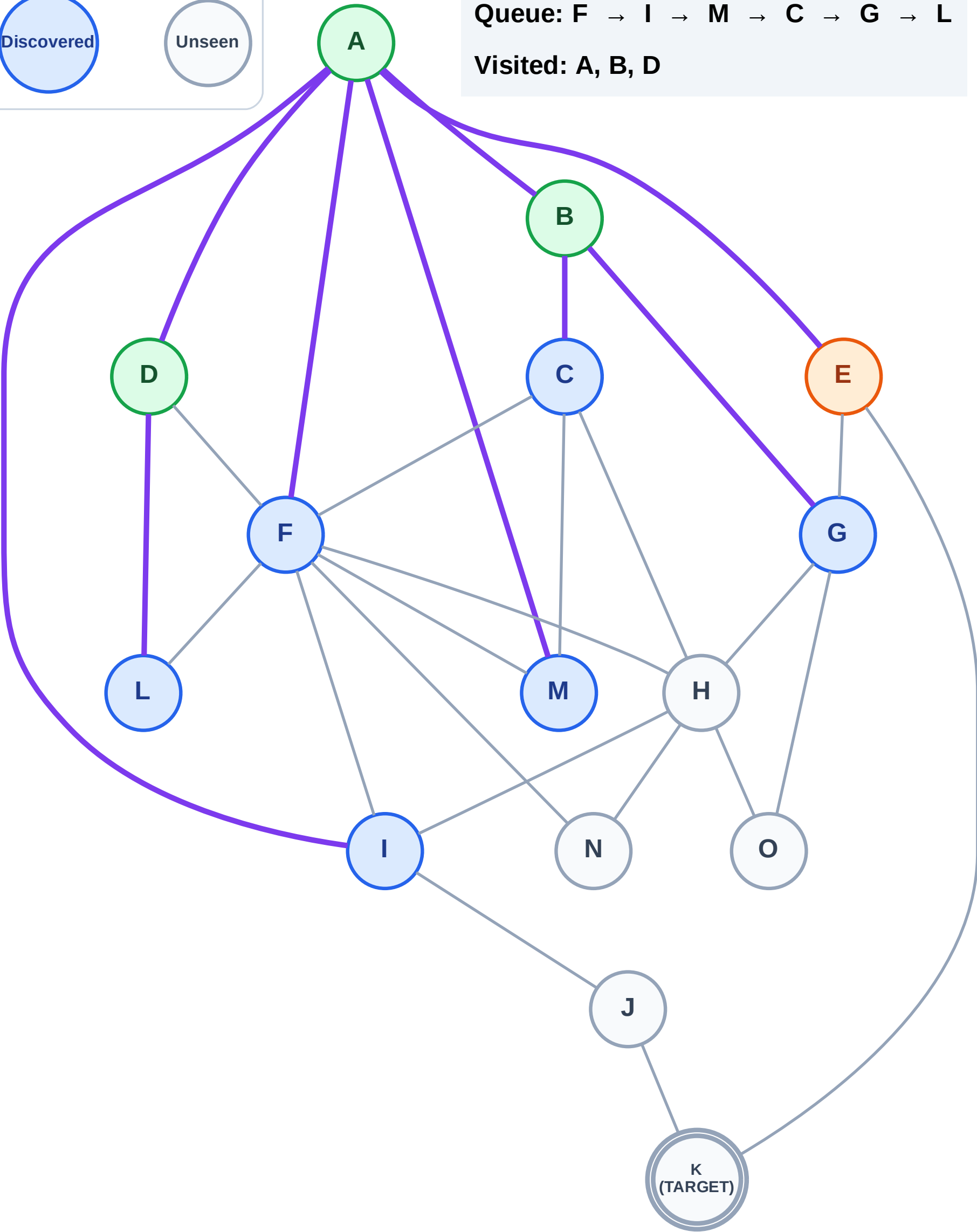
Processing

Discovered

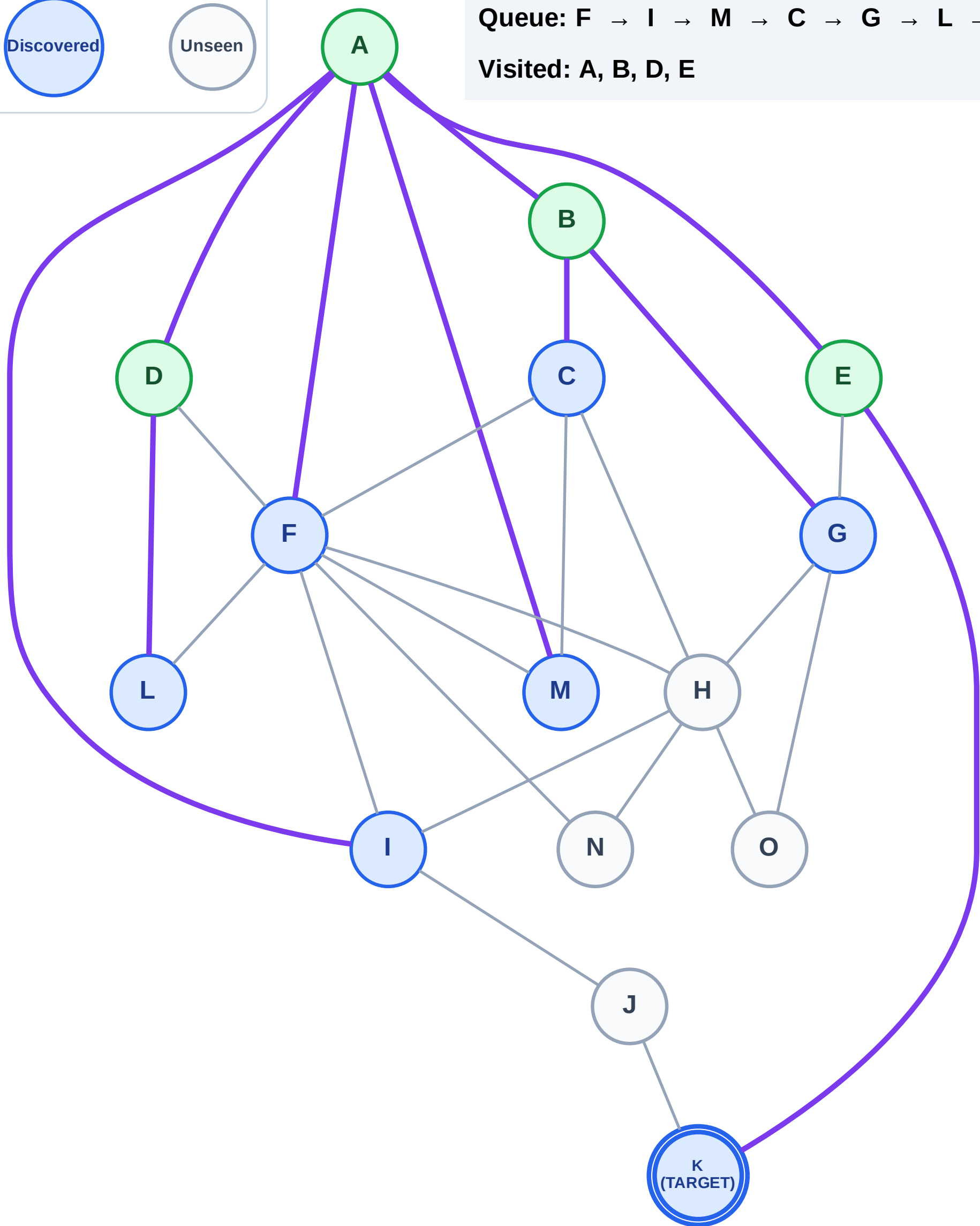
Unseen

Queue: F → I → M → C → G → L

Visited: A, B, D



Step 9: Discover K from E



BFS Traversal

Step 10: Process node F

Legend

Complete

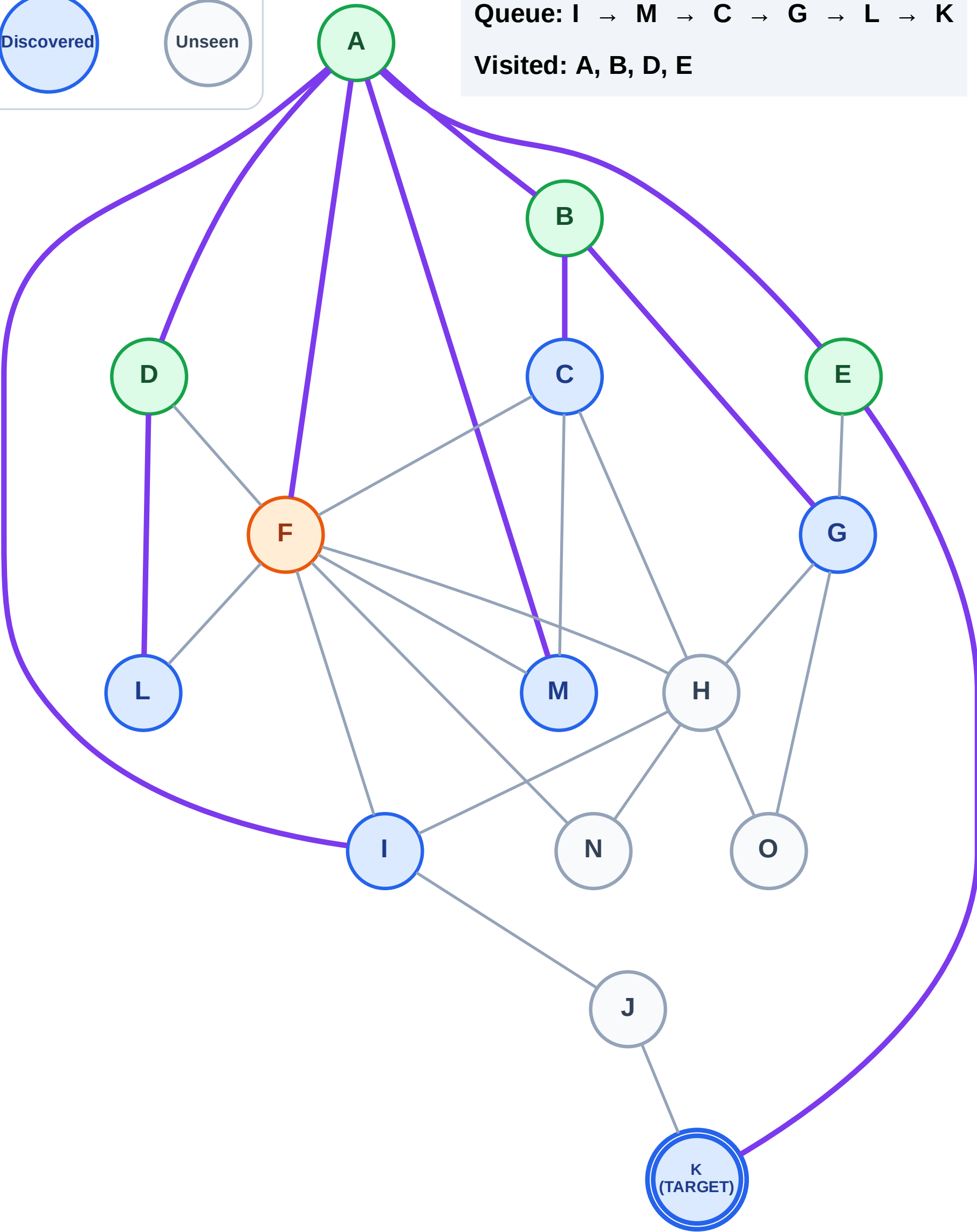
Processing

Discovered

Unseen

Queue: I → M → C → G → L → K

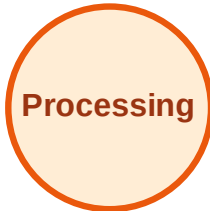
Visited: A, B, D, E



BFS Traversal

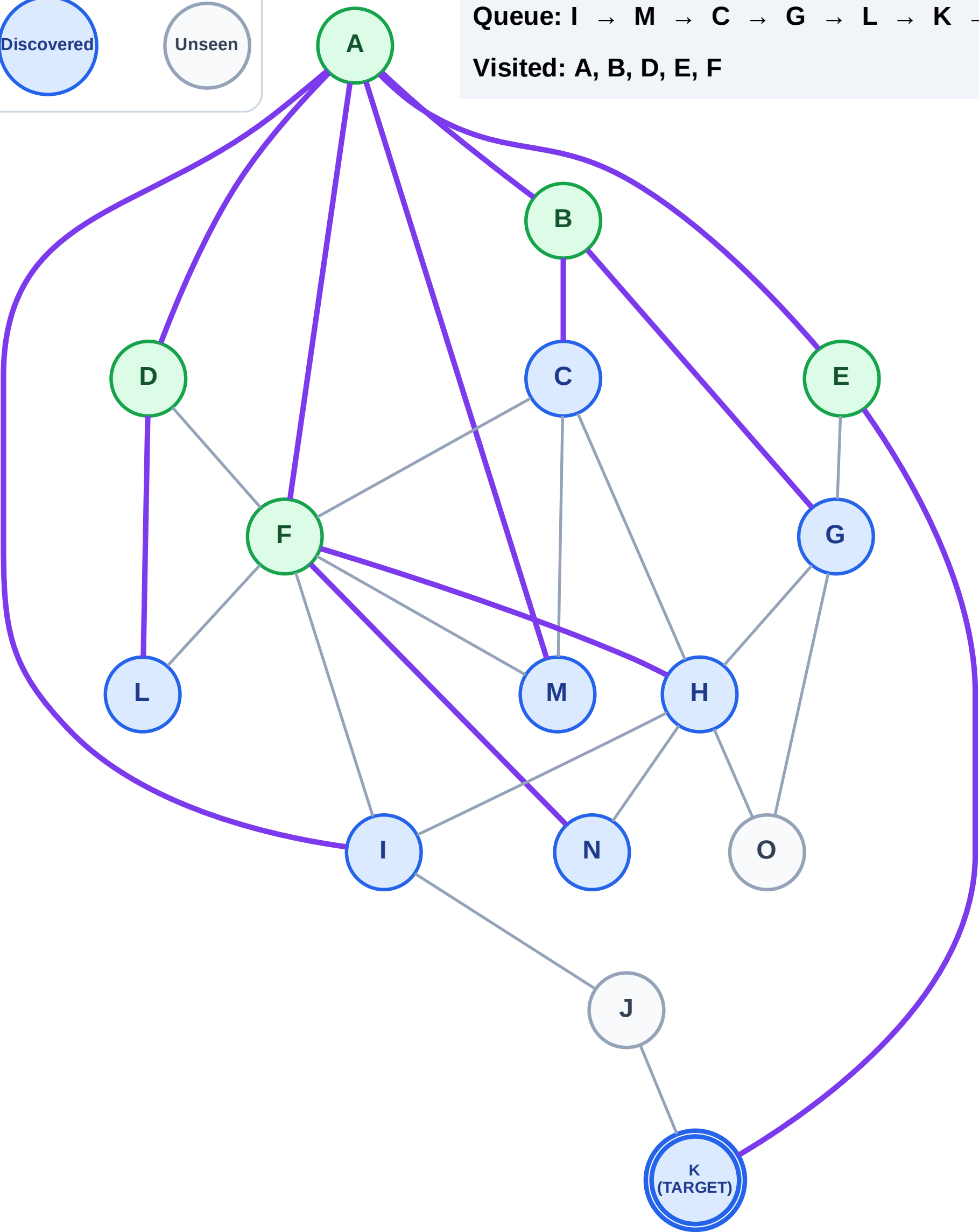
Step 11: Discover H, N from F

Legend



Queue: I → M → C → G → L → K → H → N

Visited: A, B, D, E, F



BFS Traversal

Step 12: Process node I

Legend

Complete

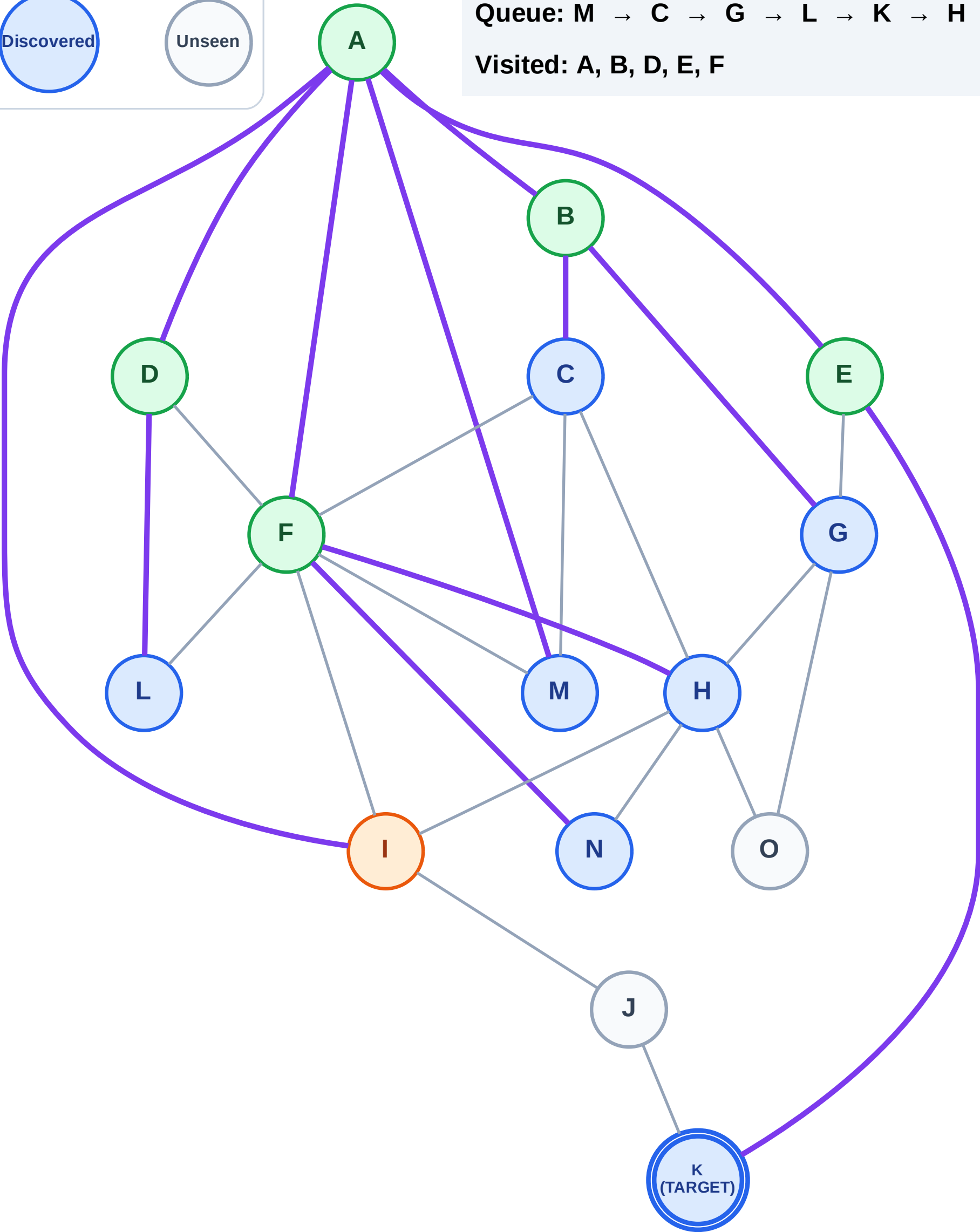
Processing

Discovered

Unseen

Queue: M → C → G → L → K → H → N

Visited: A, B, D, E, F



BFS Traversal

Step 13: Discover J from I

Legend

Complete

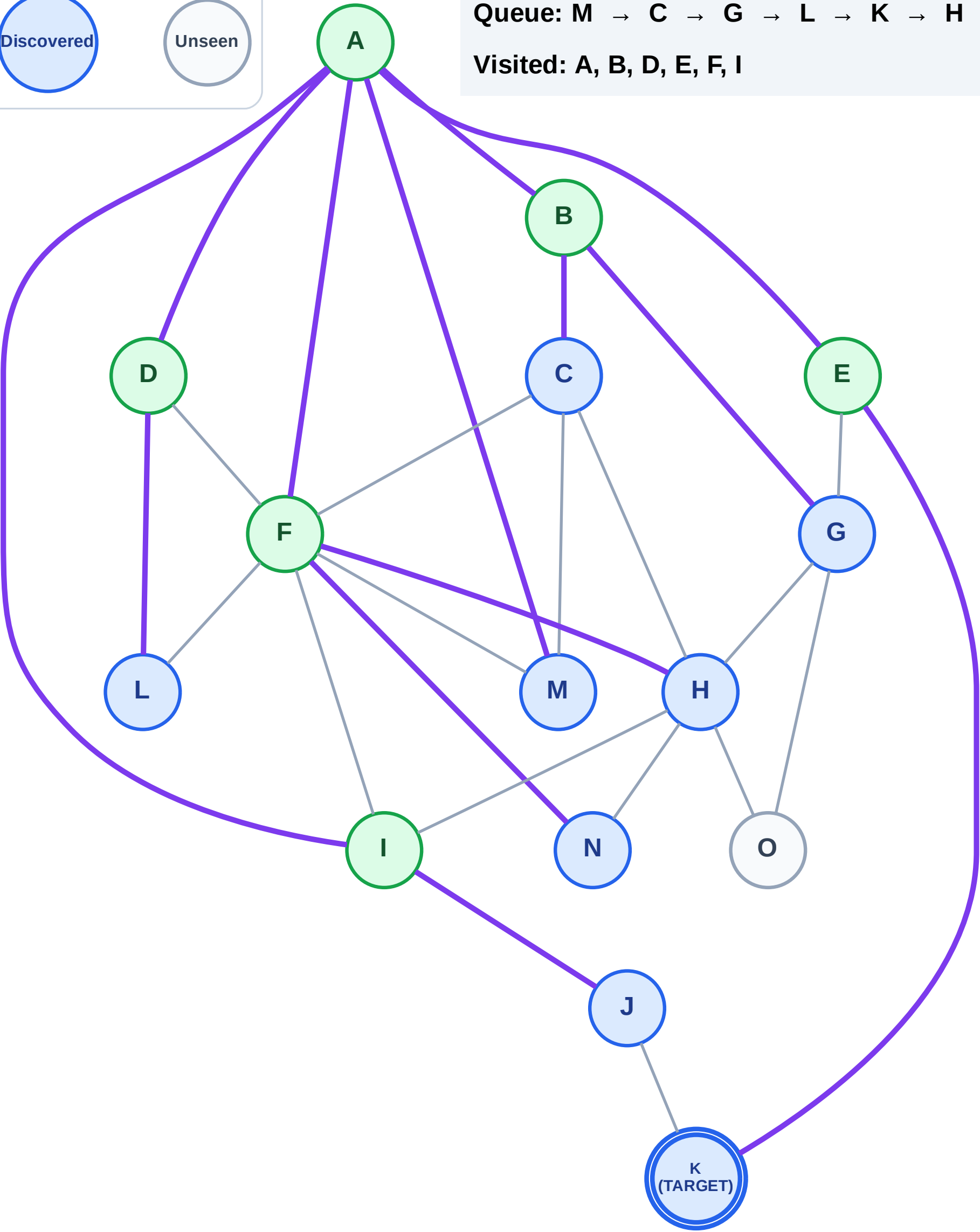
Processing

Discovered

Unseen

Queue: M → C → G → L → K → H → N → J

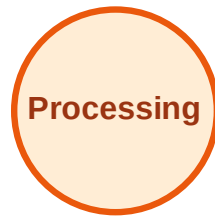
Visited: A, B, D, E, F, I



BFS Traversal

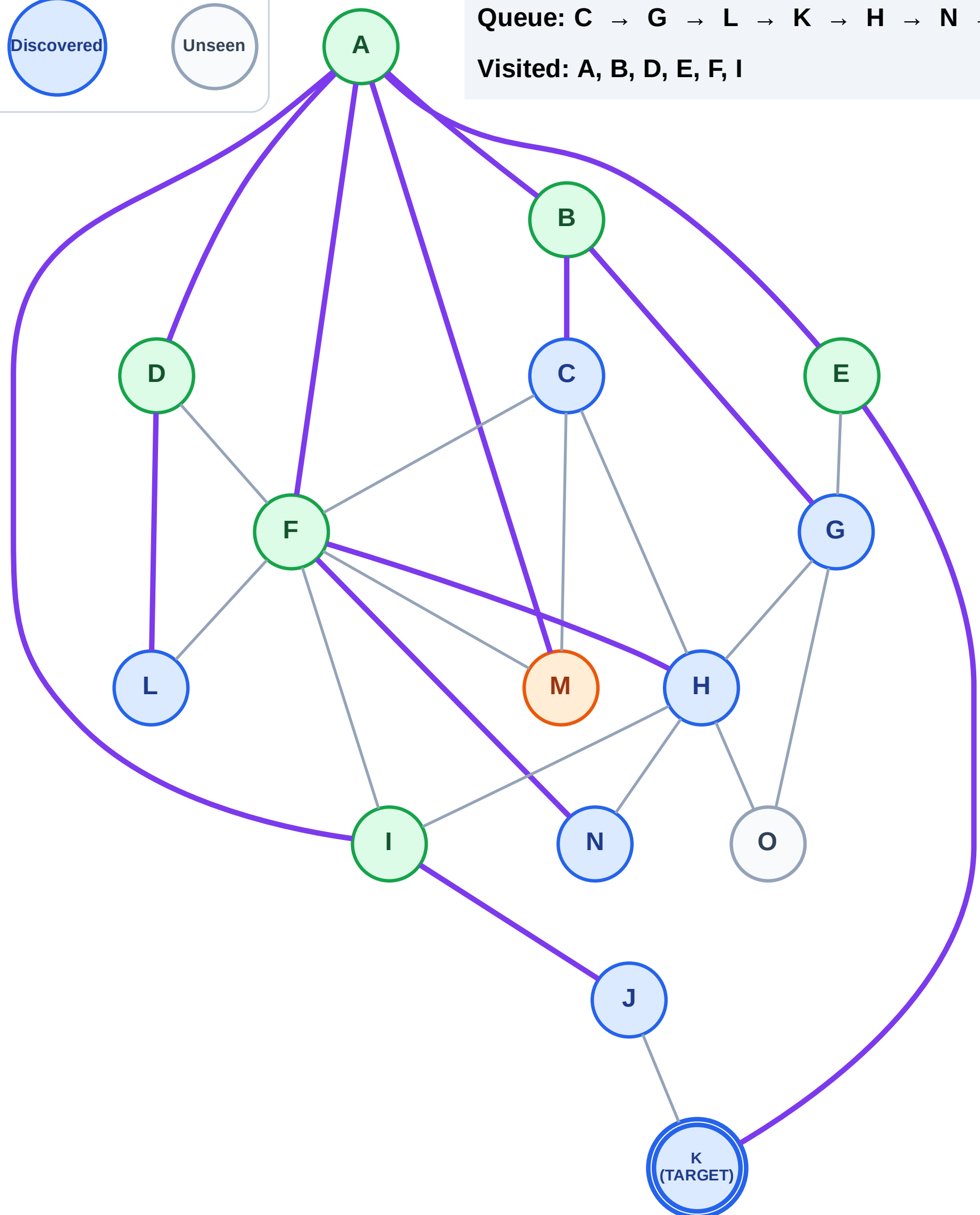
Step 14: Process node M

Legend



Queue: C → G → L → K → H → N → J

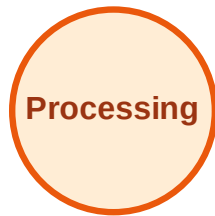
Visited: A, B, D, E, F, I



BFS Traversal

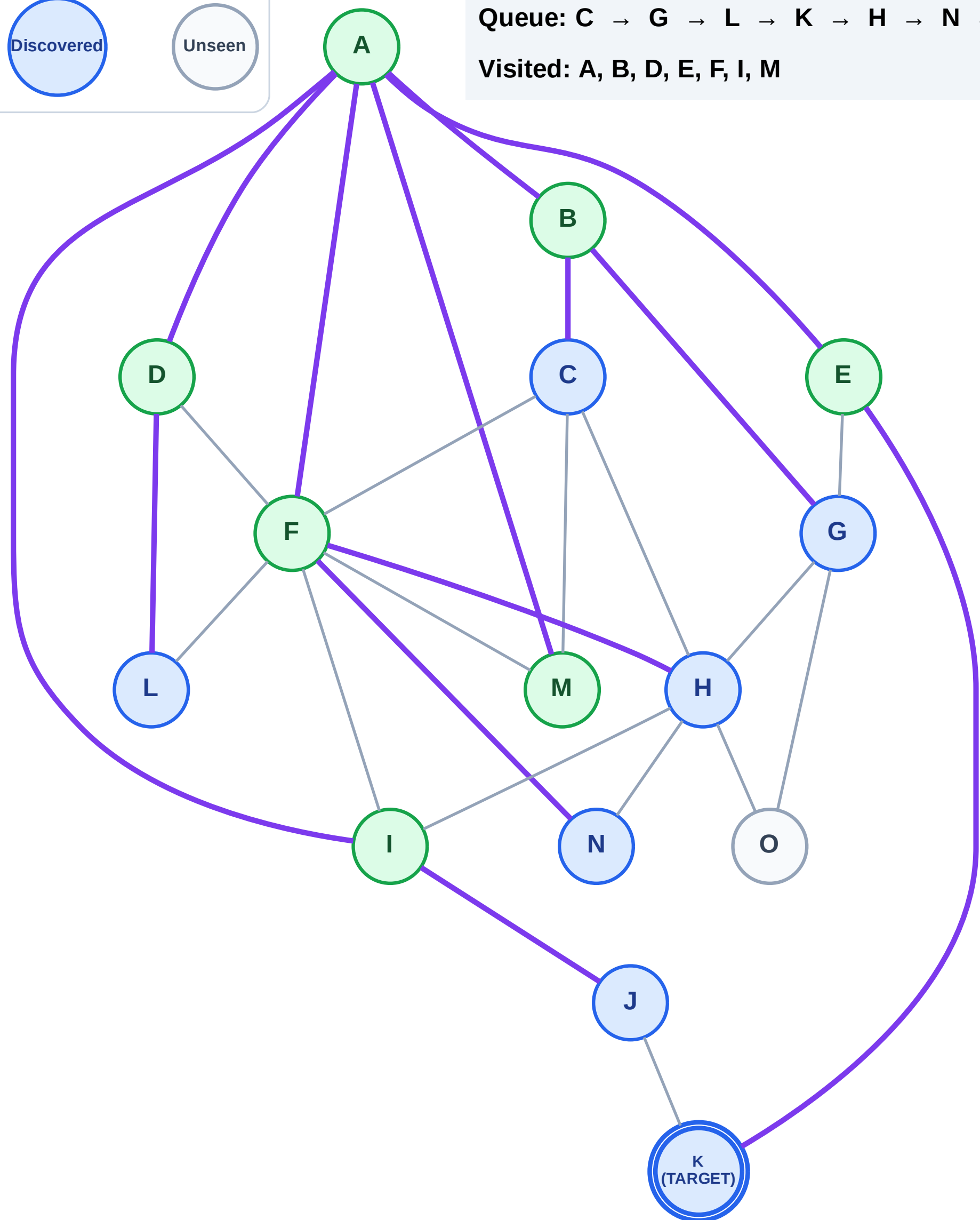
Step 15: No new neighbors from M

Legend



Queue: C → G → L → K → H → N → J

Visited: A, B, D, E, F, I, M



BFS Traversal

Step 16: Process node C

Legend

Complete

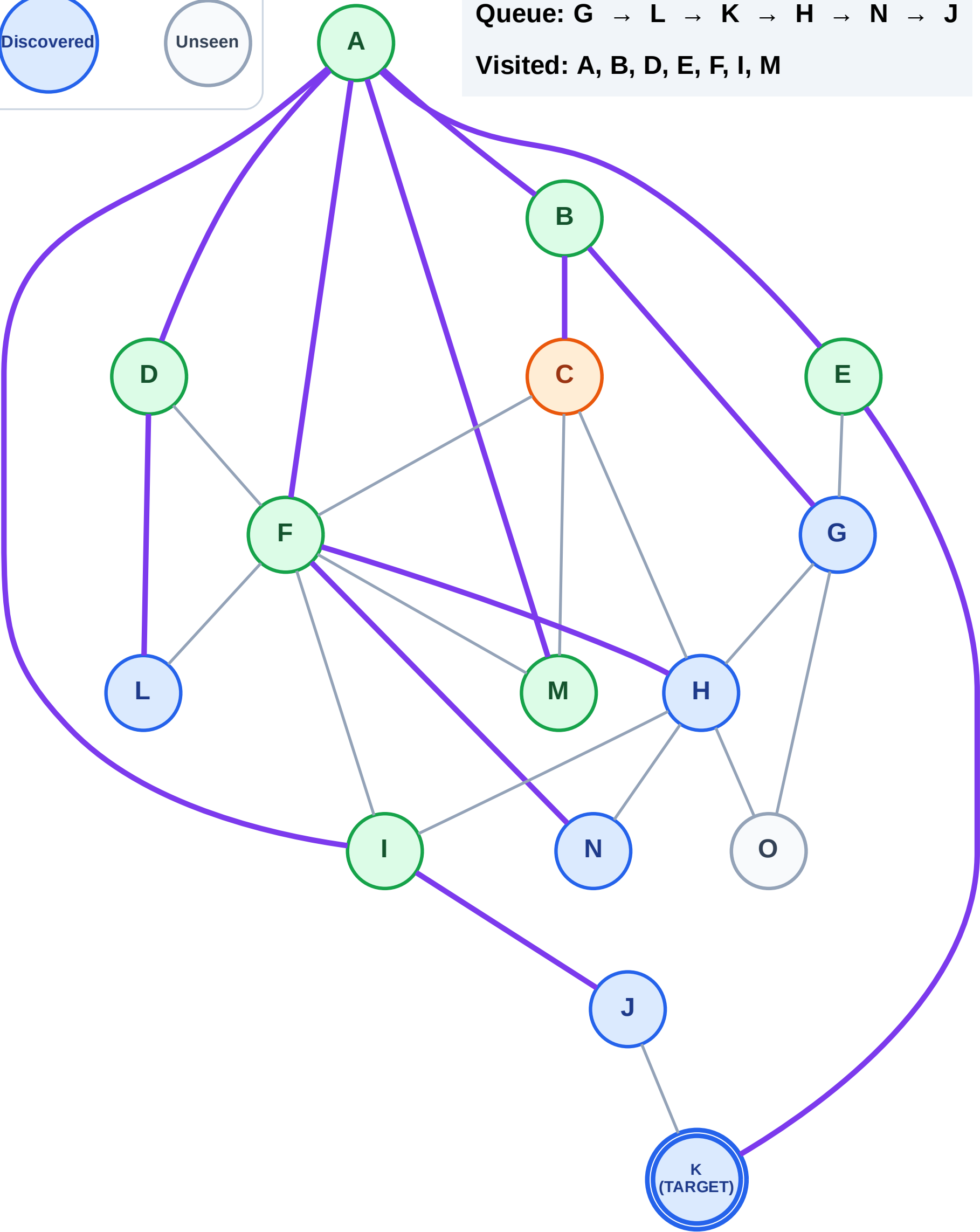
Processing

Discovered

Unseen

Queue: G → L → K → H → N → J

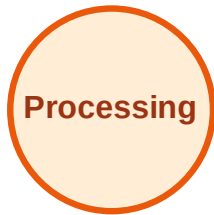
Visited: A, B, D, E, F, I, M



BFS Traversal

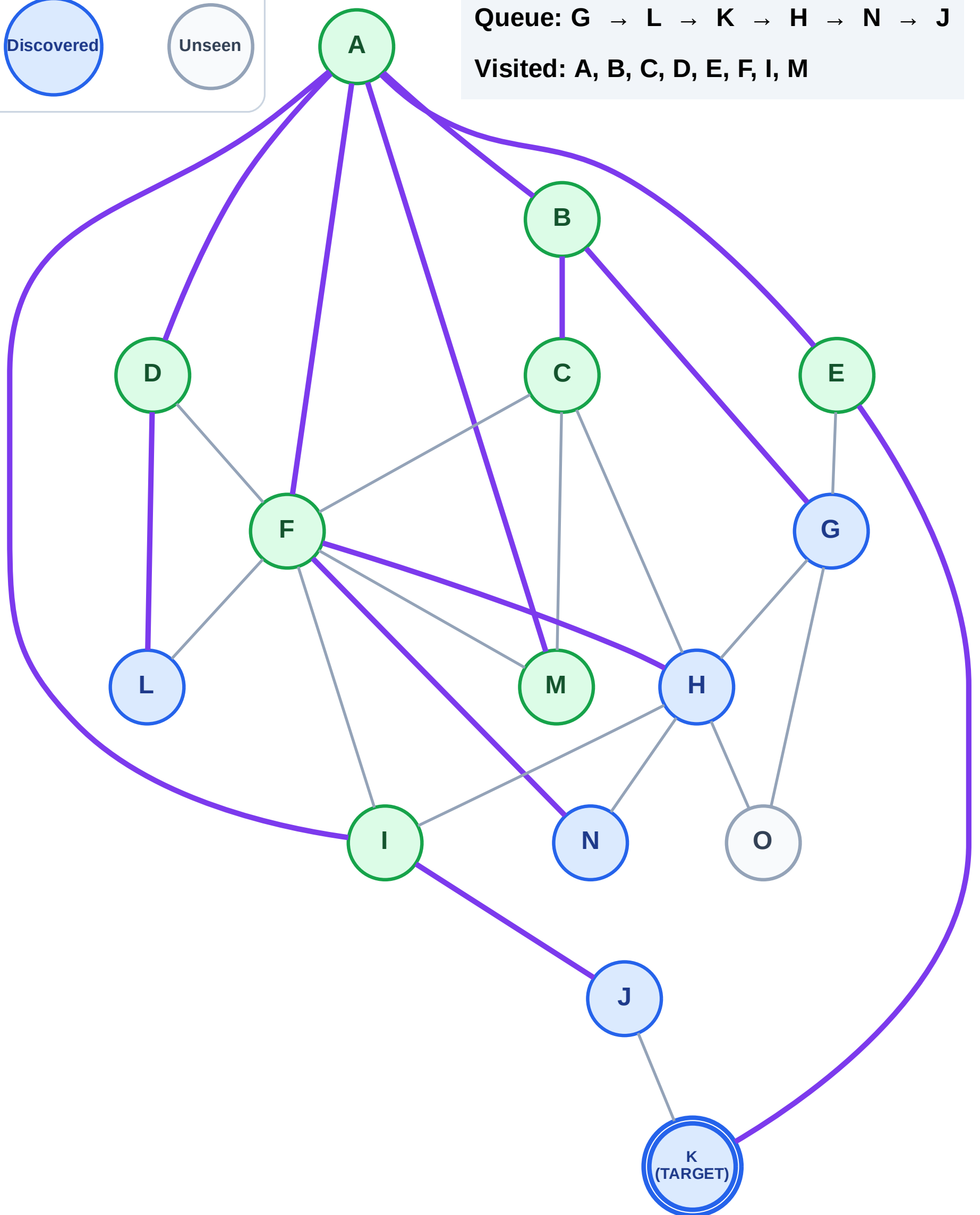
Step 17: No new neighbors from C

Legend



Queue: G → L → K → H → N → J

Visited: A, B, C, D, E, F, I, M



BFS Traversal

Step 18: Process node G

Legend

Complete

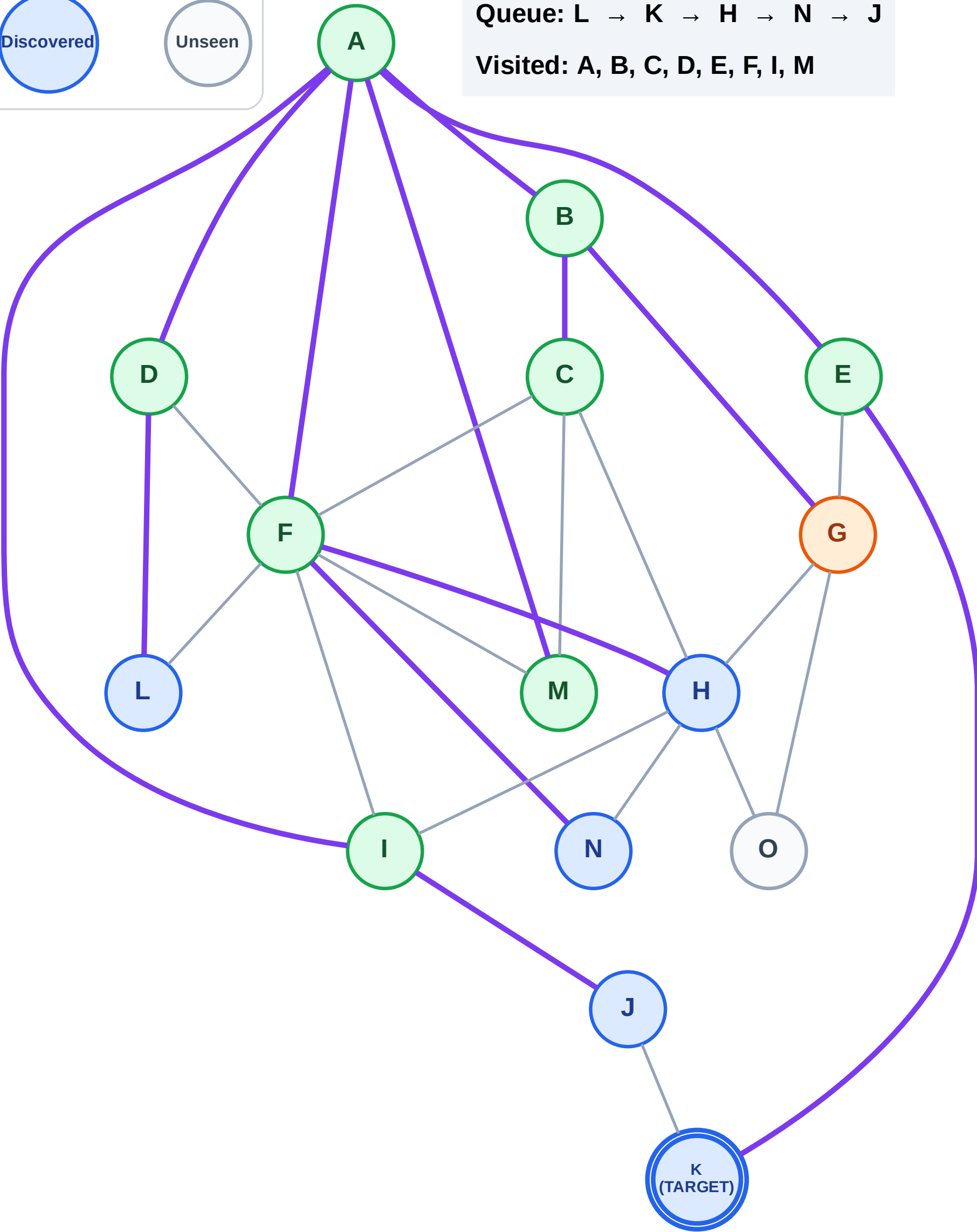
Processing

Discovered

Unseen

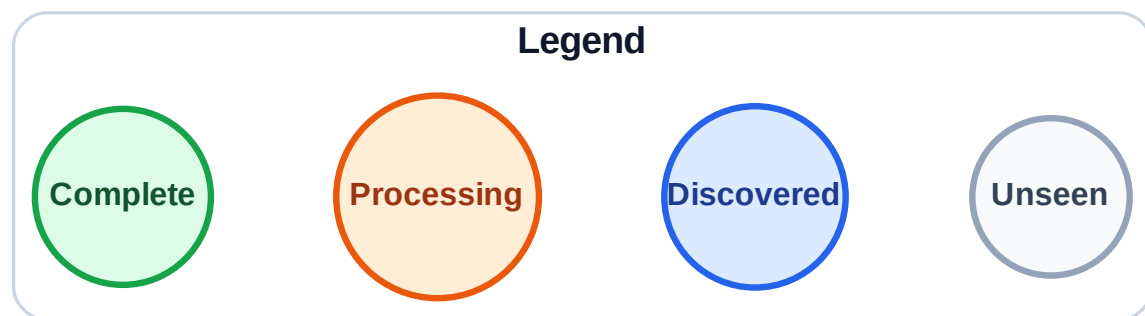
Queue: L → K → H → N → J

Visited: A, B, C, D, E, F, I, M



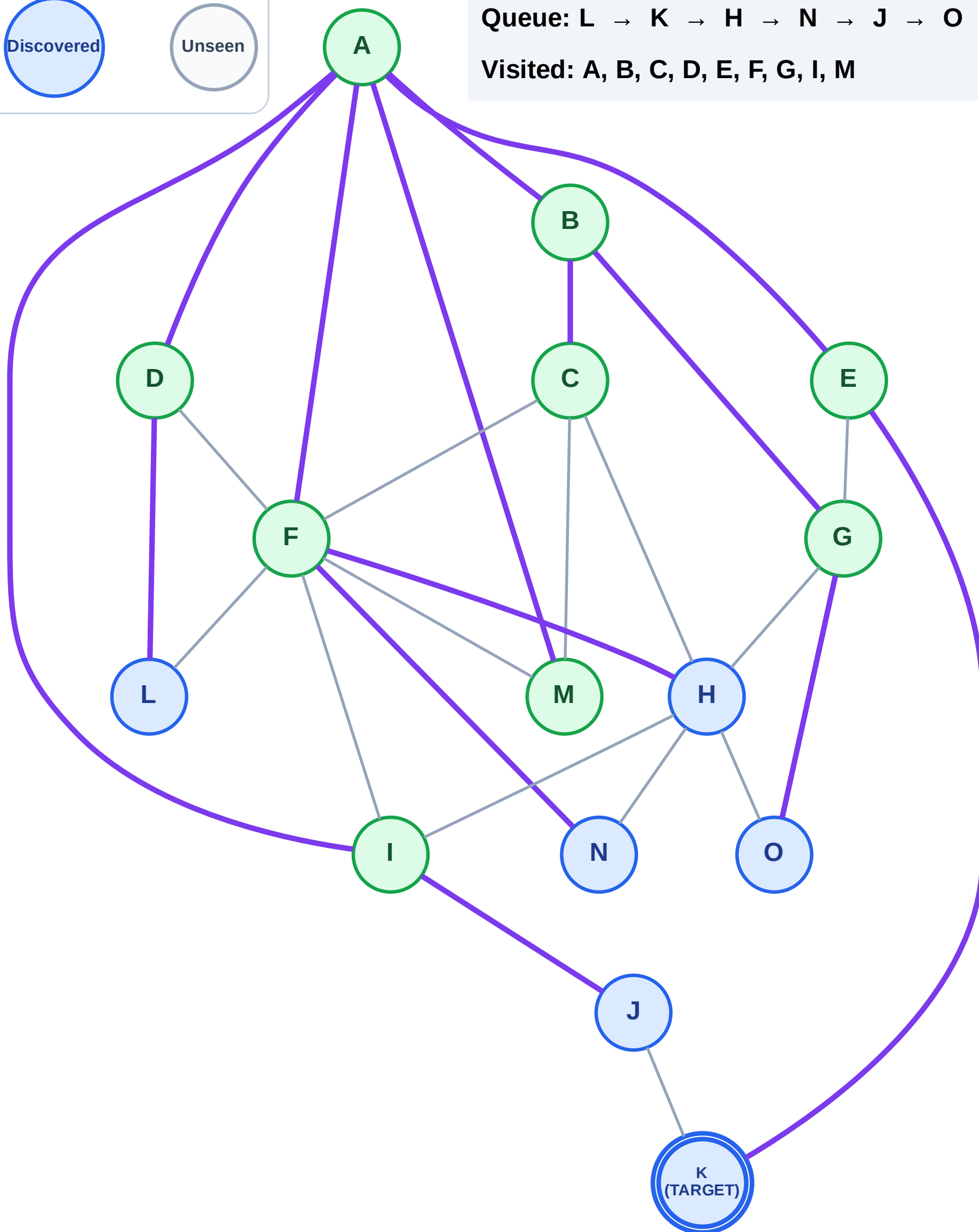
BFS Traversal

Step 19: Discover O from G



Queue: L → K → H → N → J → O

Visited: A, B, C, D, E, F, G, I, M



BFS Traversal

Step 20: Process node L

Legend

Complete

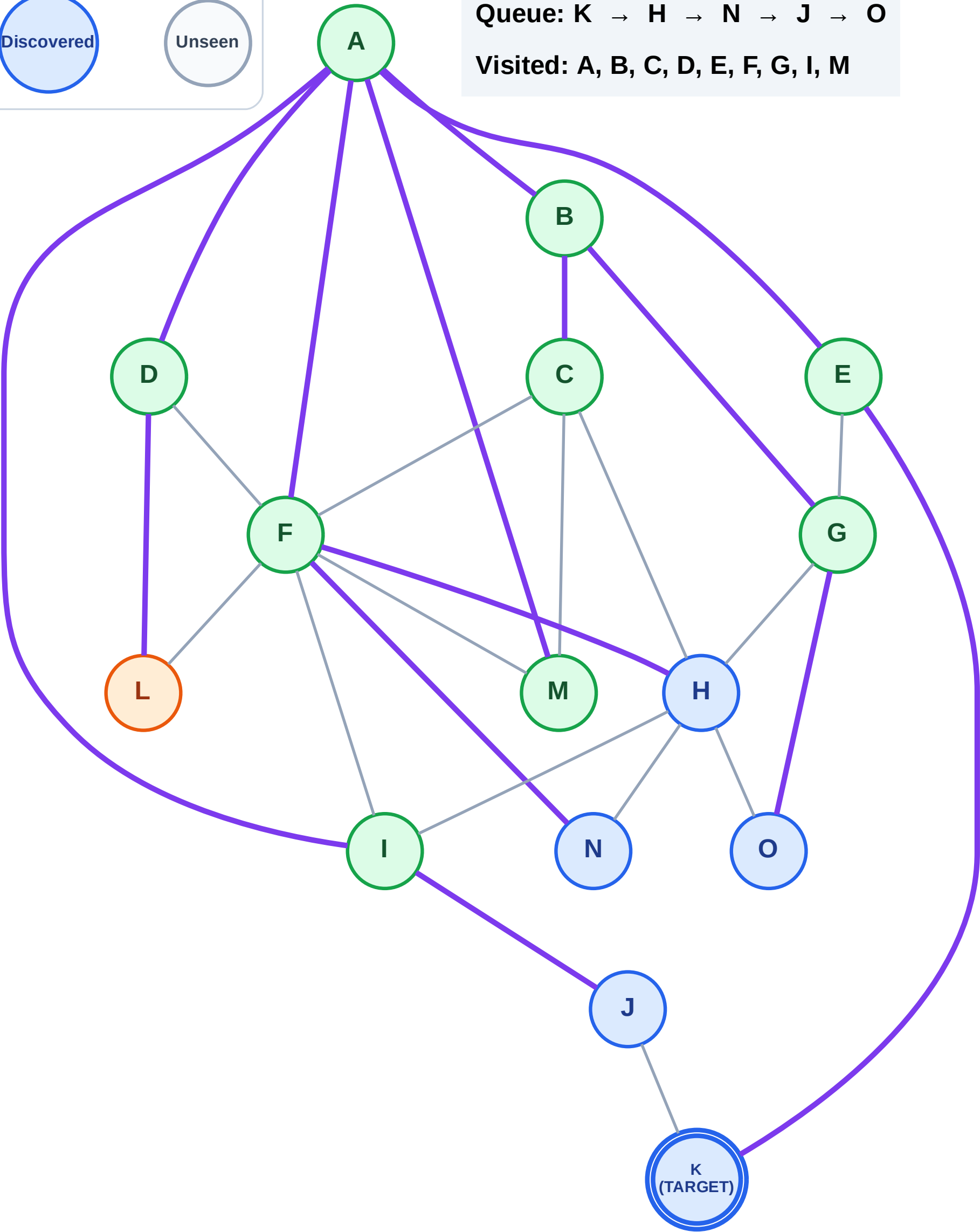
Processing

Discovered

Unseen

Queue: K → H → N → J → O

Visited: A, B, C, D, E, F, G, I, M



BFS Traversal

Step 21: No new neighbors from L

Legend

Complete

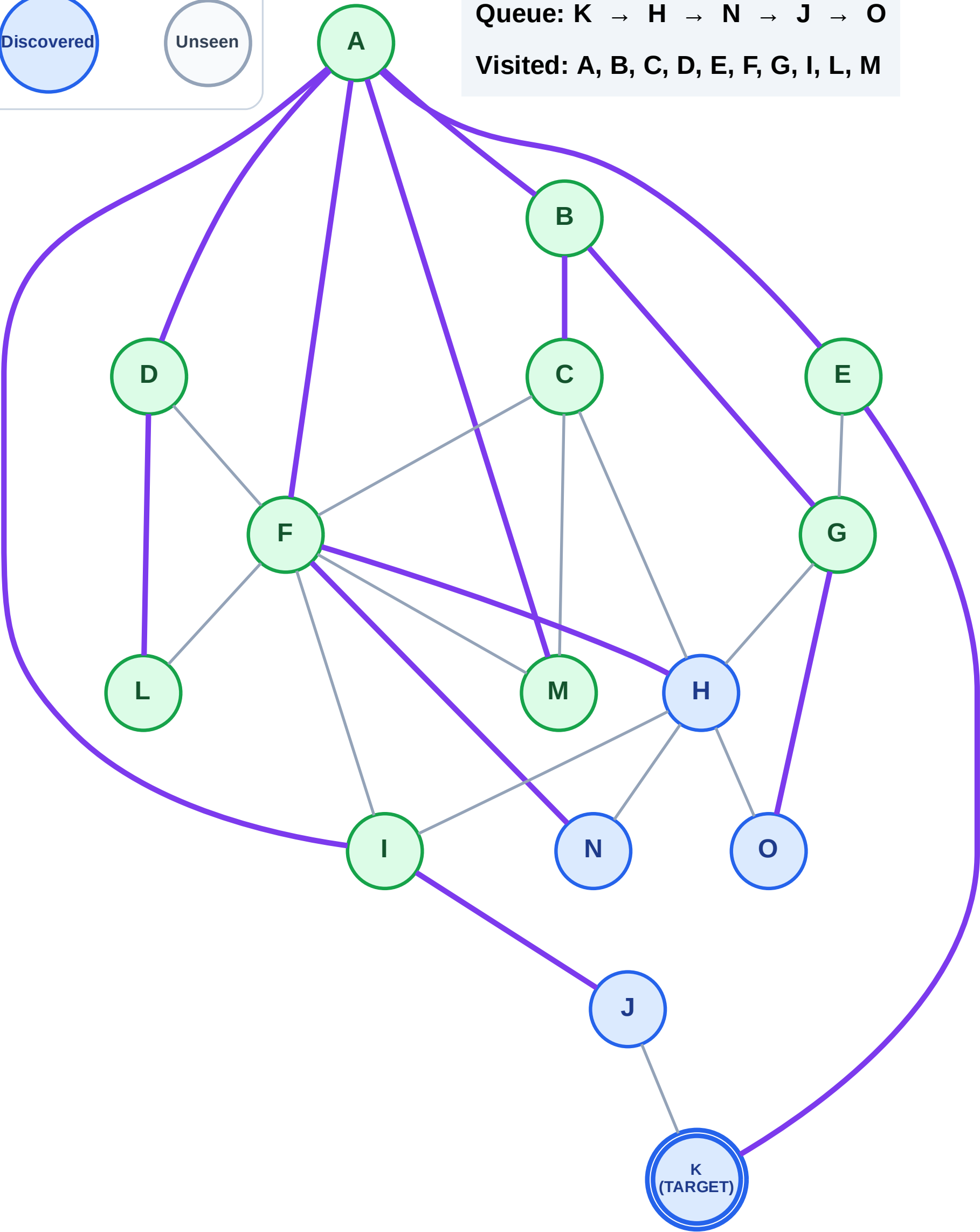
Processing

Discovered

Unseen

Queue: K → H → N → J → O

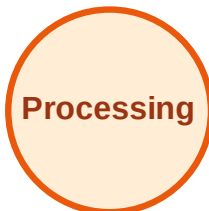
Visited: A, B, C, D, E, F, G, I, L, M



BFS Traversal

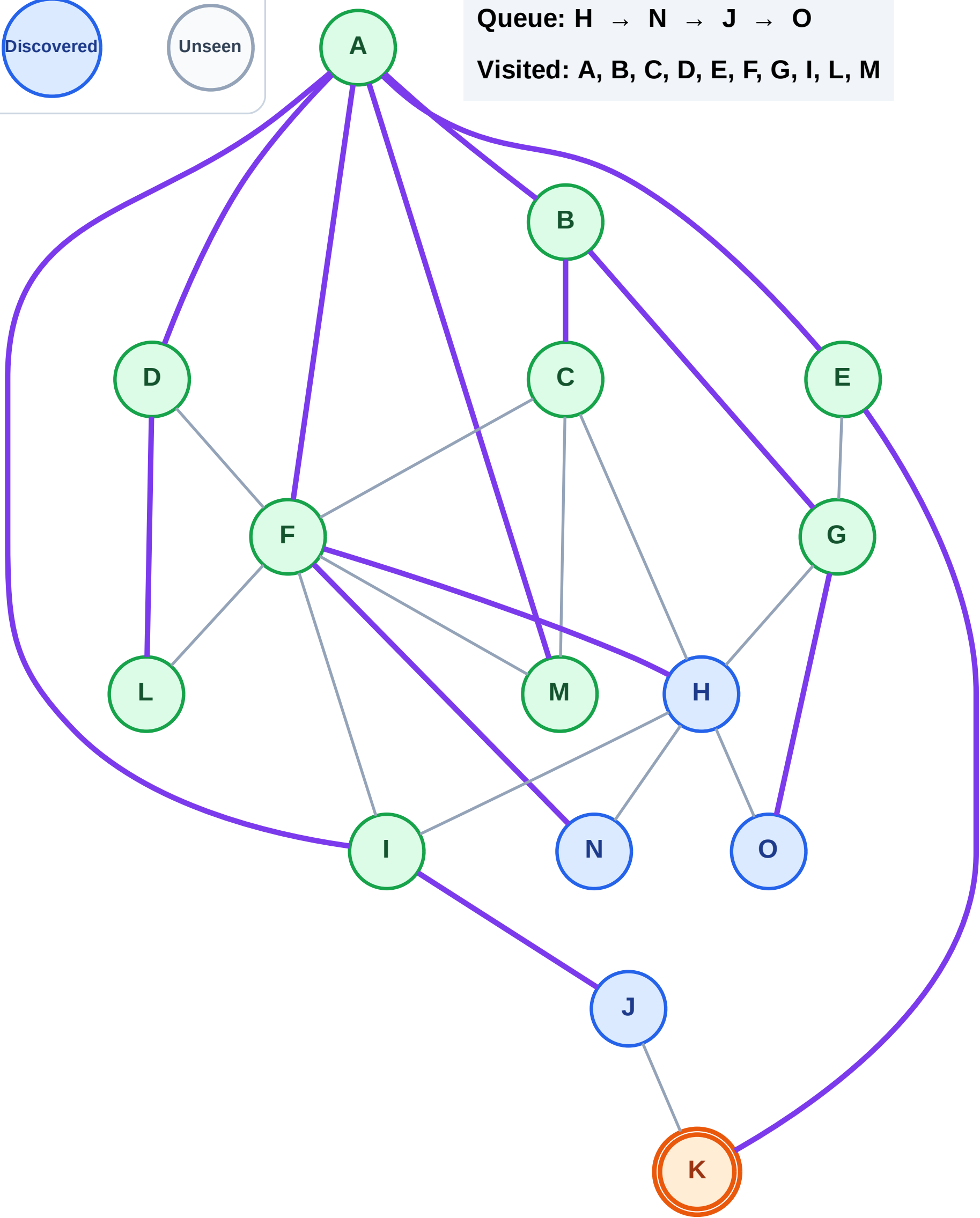
Step 22: Process node K

Legend



Queue: H → N → J → O

Visited: A, B, C, D, E, F, G, I, L, M



BFS Traversal

Step 23: TARGET NODE K REACHED! Path (2 hops): A → E → K

Legend

Complete

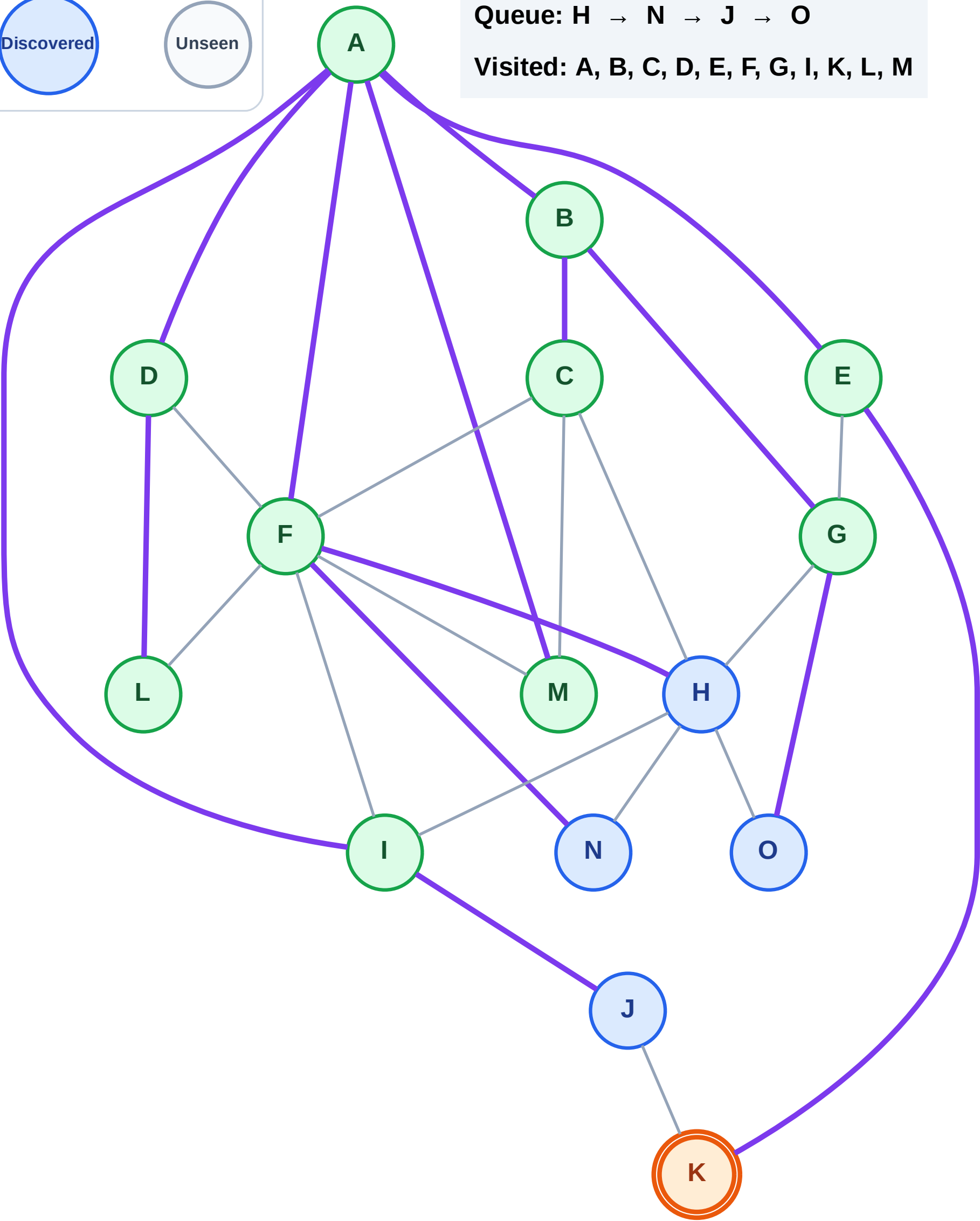
Processing

Discovered

Unseen

Queue: H → N → J → O

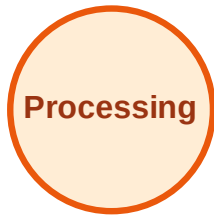
Visited: A, B, C, D, E, F, G, I, K, L, M



DFS Traversal

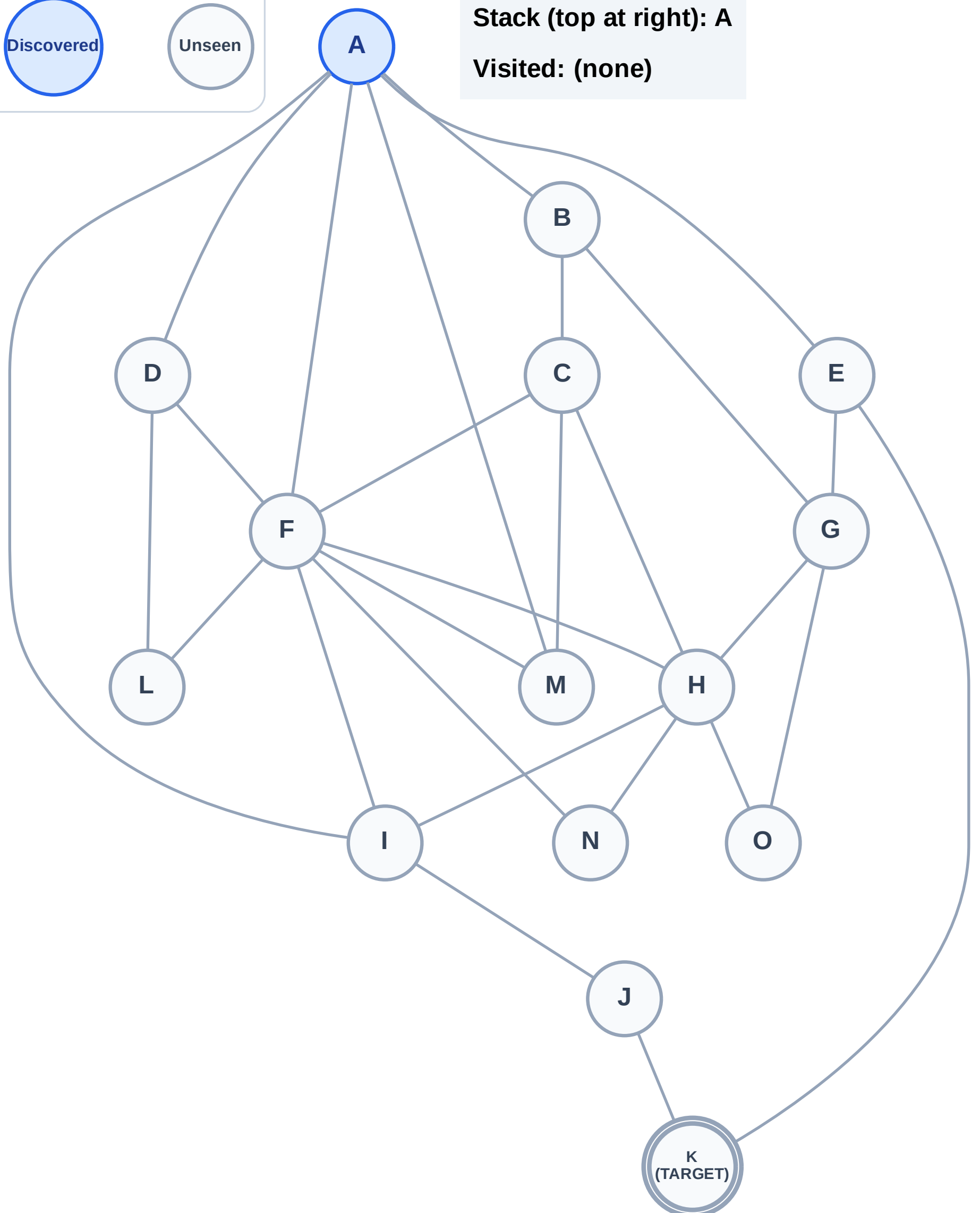
Step 1: Start at A (Seeking target node K)

Legend



Stack (top at right): A

Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

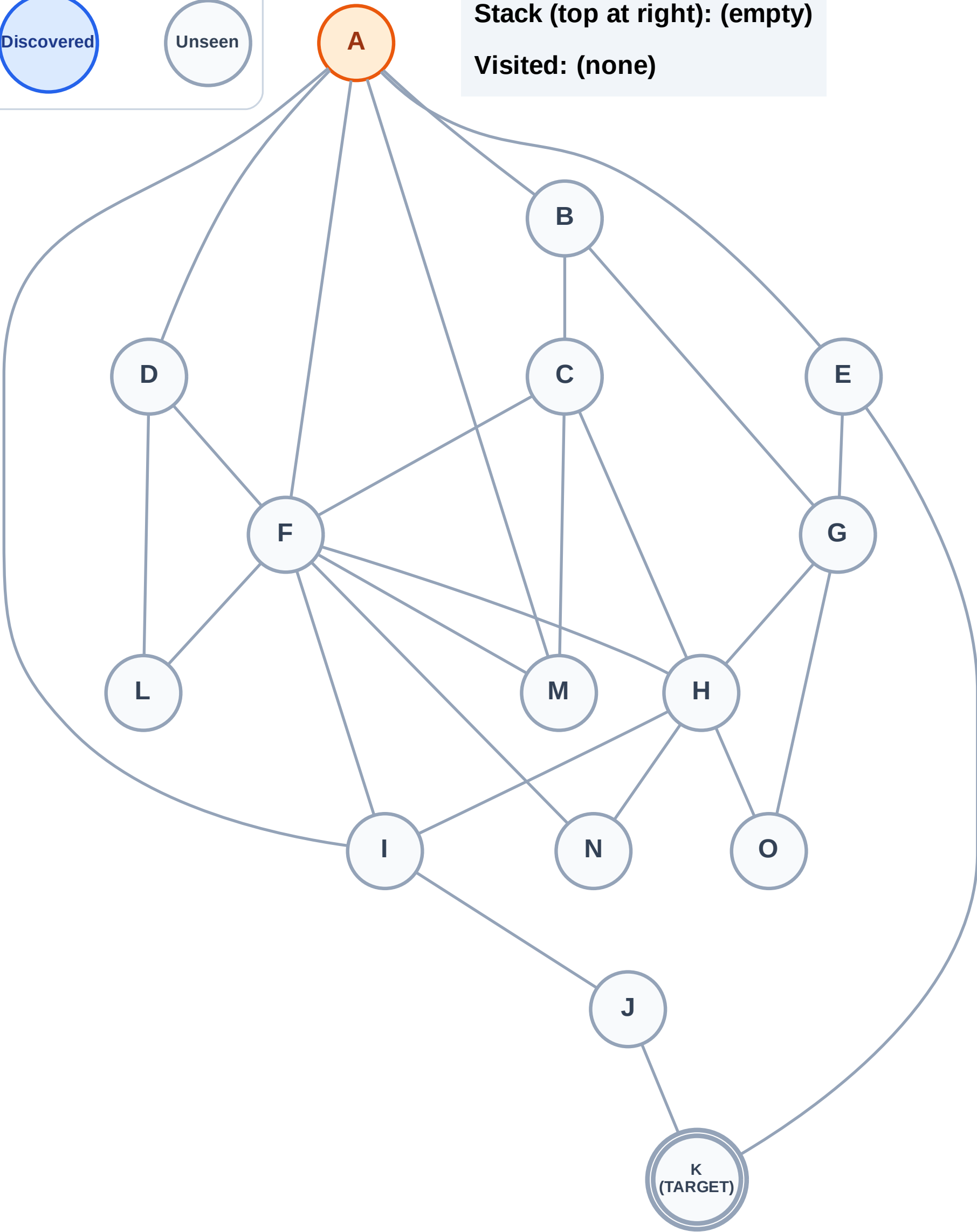
Complete

Processing

Discovered

Unseen

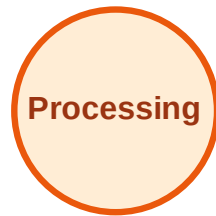
Stack (top at right): (empty)
Visited: (none)



DFS Traversal

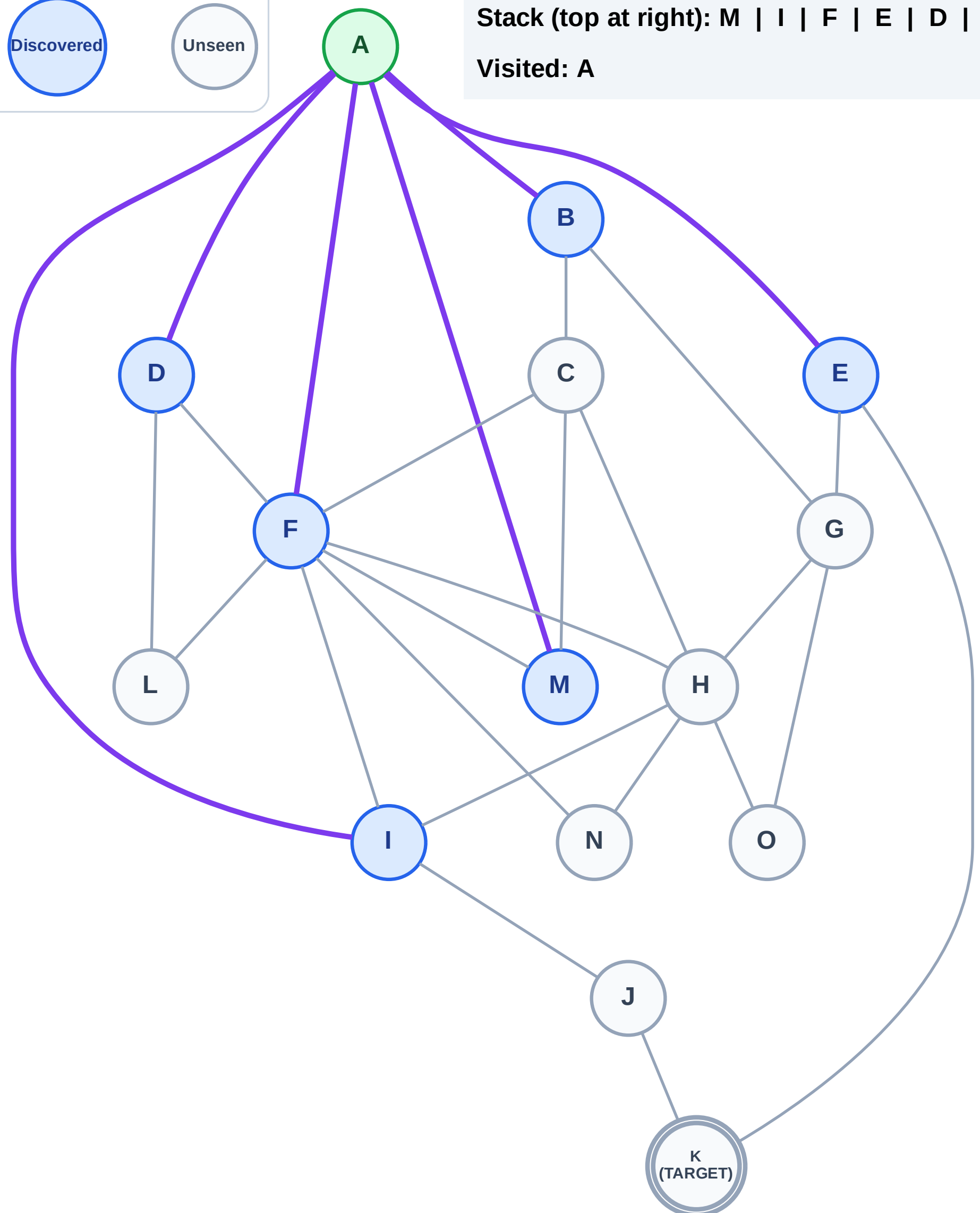
Step 3: Discover M, I, F, E, D, B from A

Legend



Stack (top at right): M | I | F | E | D | B

Visited: A



DFS Traversal

Step 4: Process node B

Legend

Complete

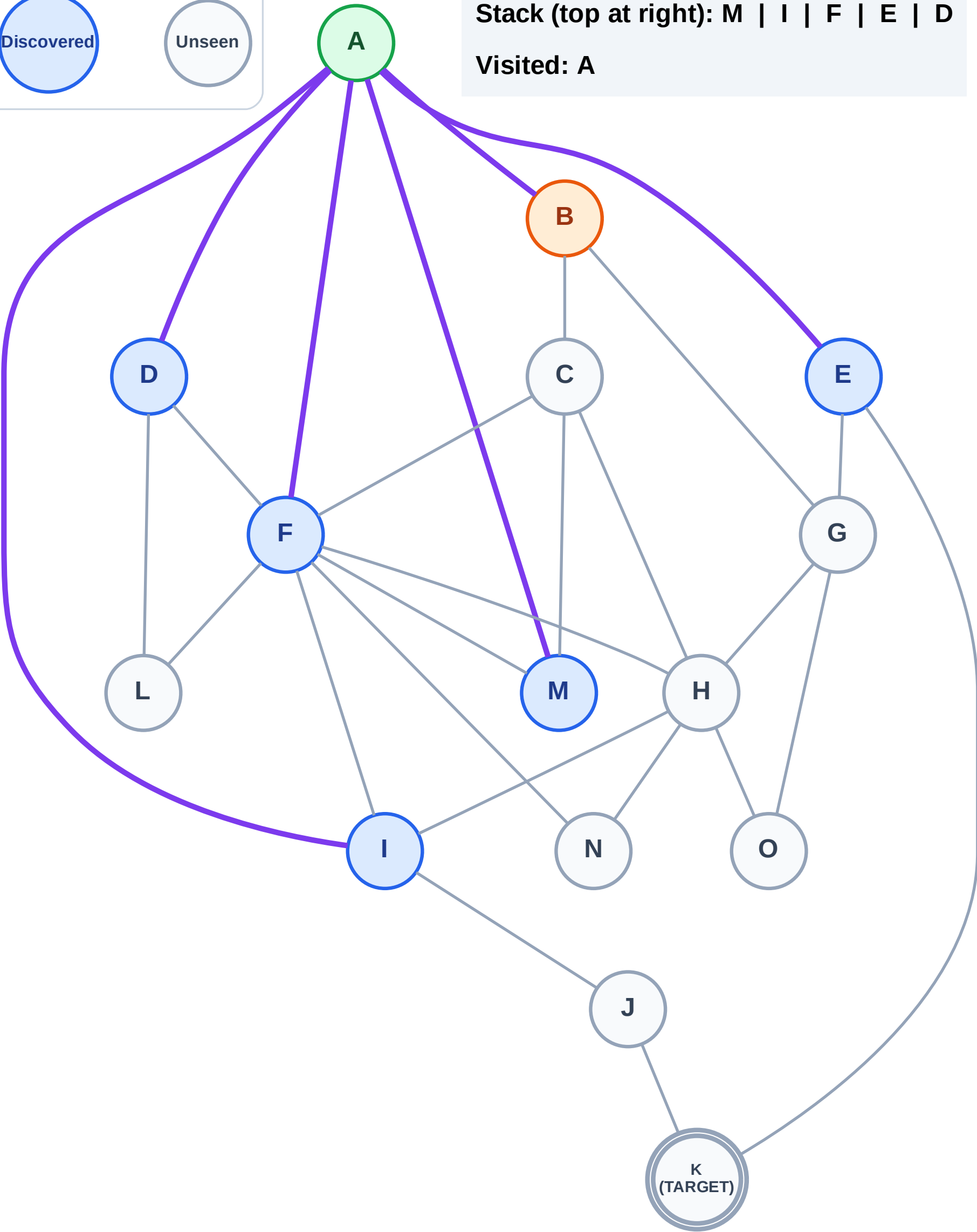
Processing

Discovered

Unseen

Stack (top at right): M | I | F | E | D

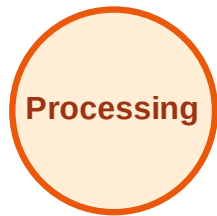
Visited: A



DFS Traversal

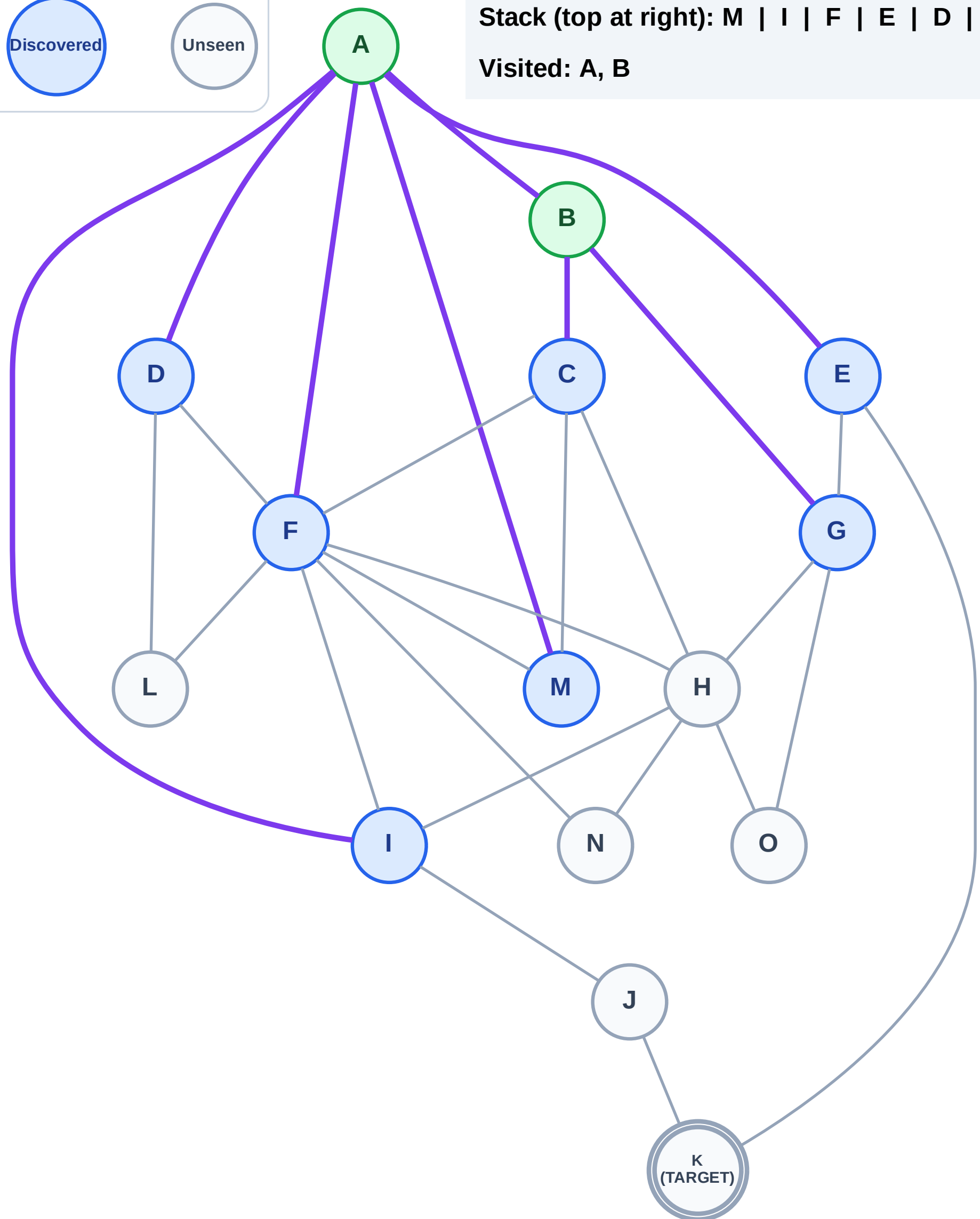
Step 5: Discover G, C from B

Legend



Stack (top at right): M | I | F | E | D | G | C

Visited: A, B



DFS Traversal

Step 6: Process node C

Legend

Complete

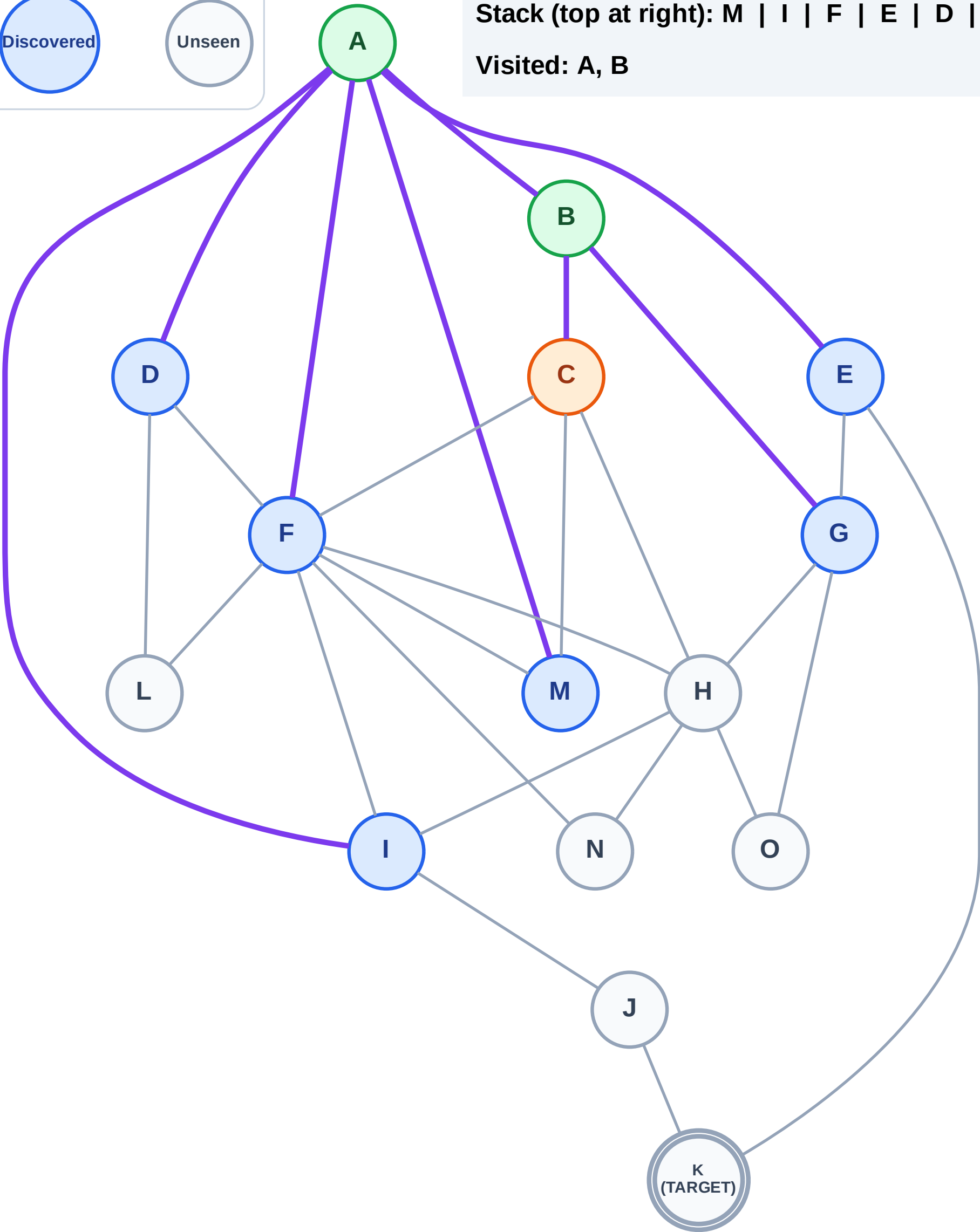
Processing

Discovered

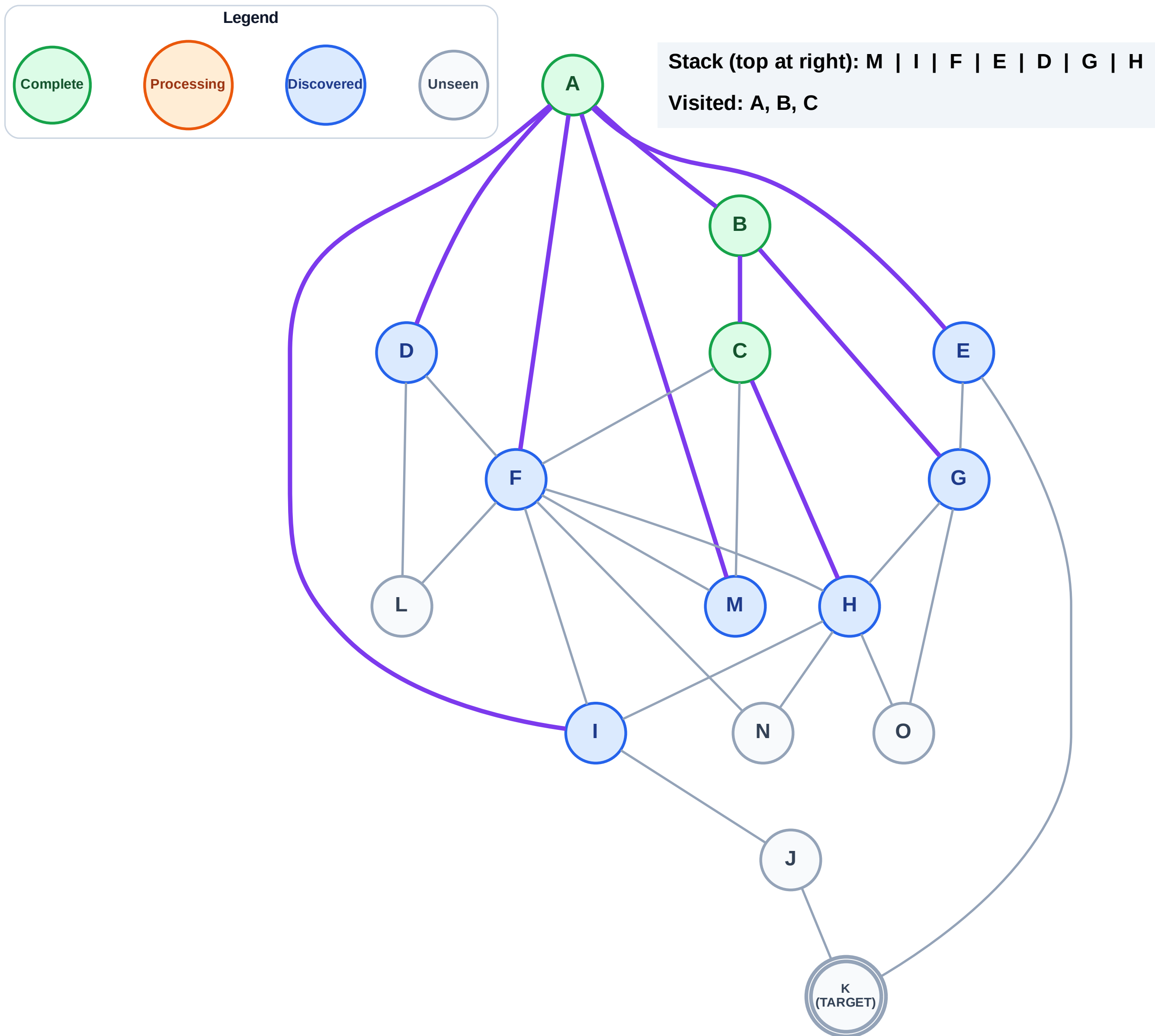
Unseen

Stack (top at right): M | I | F | E | D | G

Visited: A, B



Step 7: Discover H from C



DFS Traversal

Step 8: Process node H

Legend

Complete

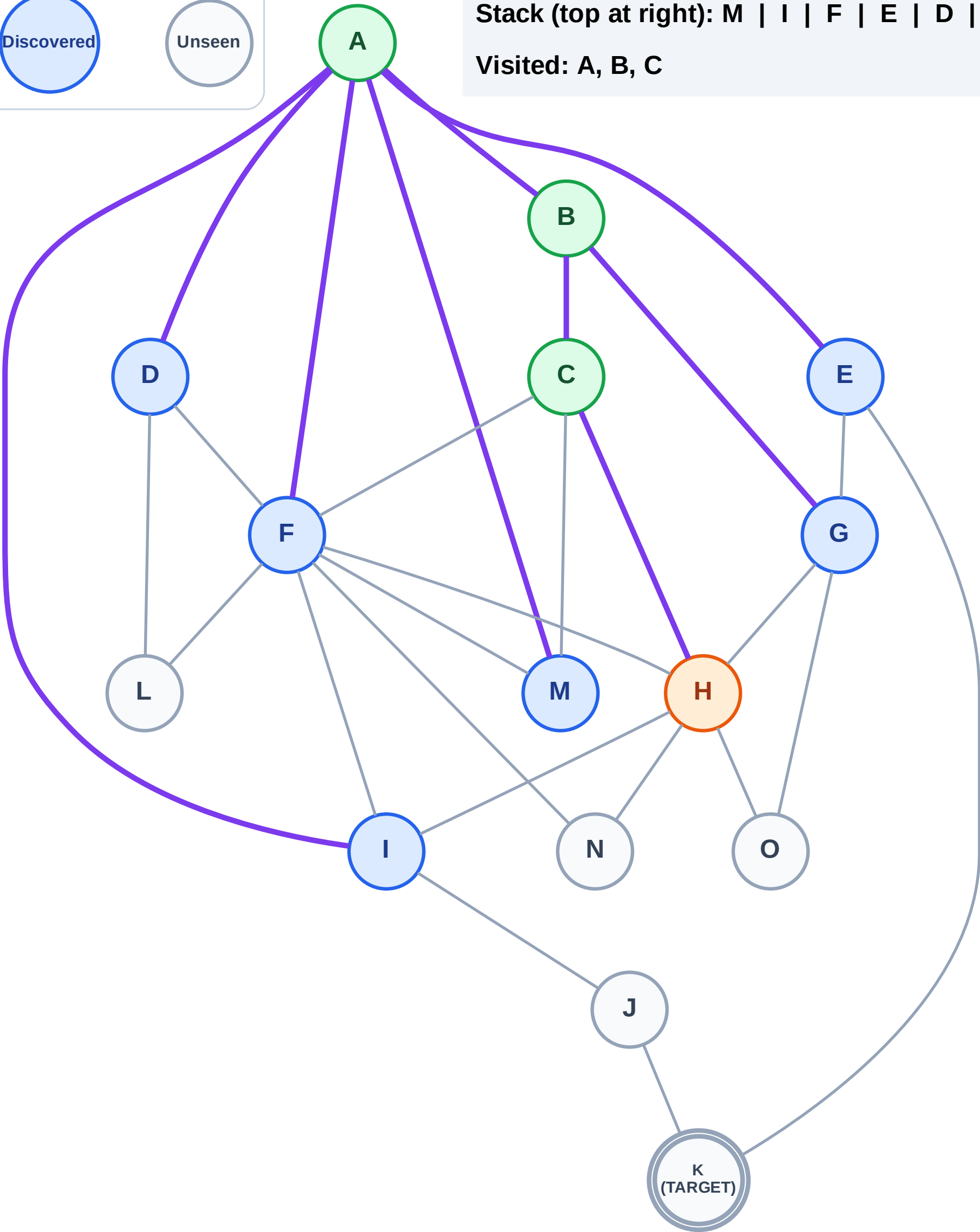
Processing

Discovered

Unseen

Stack (top at right): M | I | F | E | D | G

Visited: A, B, C



DFS Traversal

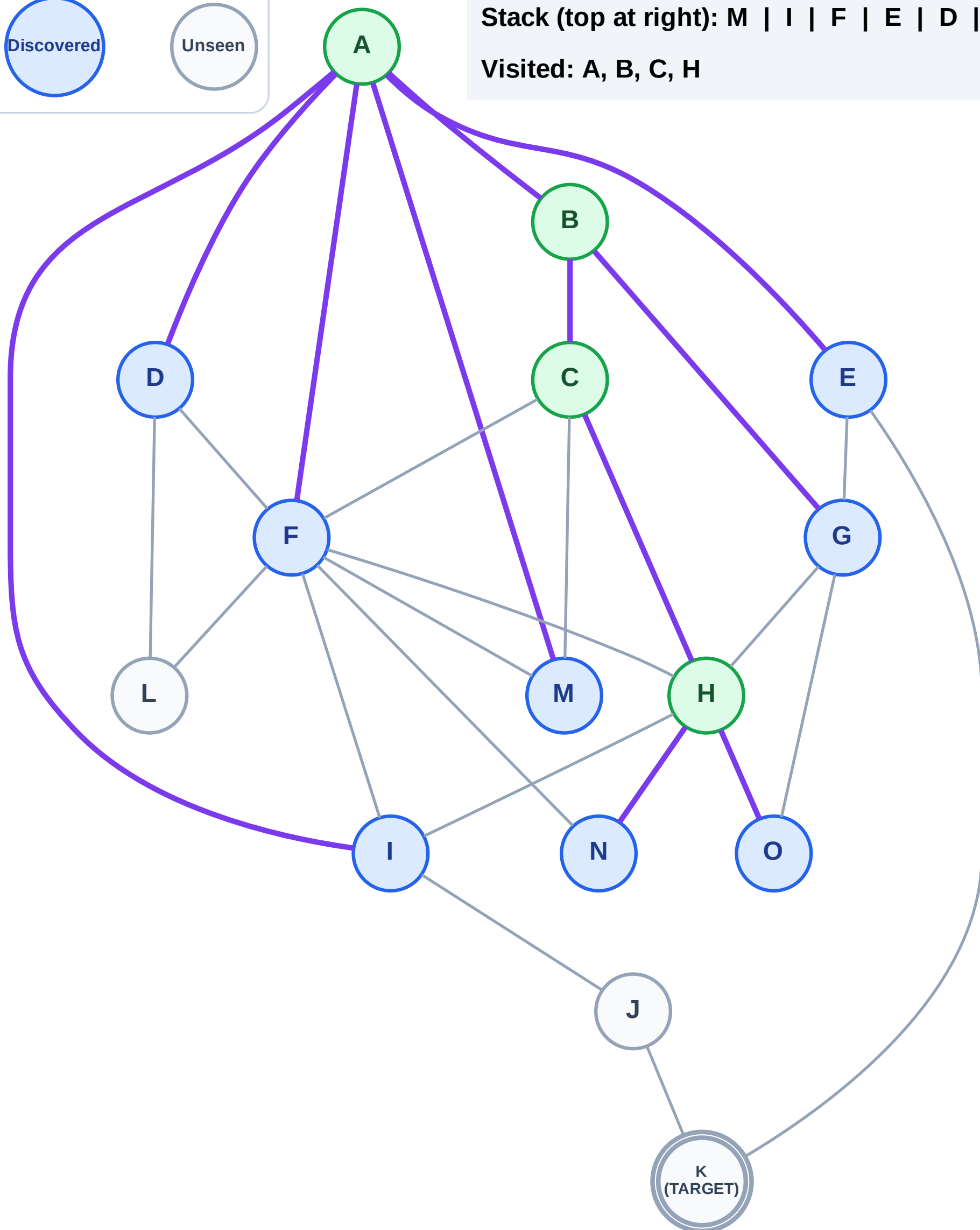
Step 9: Discover O, N from H

Legend



Stack (top at right): M | I | F | E | D | G | O | N

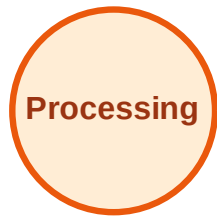
Visited: A, B, C, H



DFS Traversal

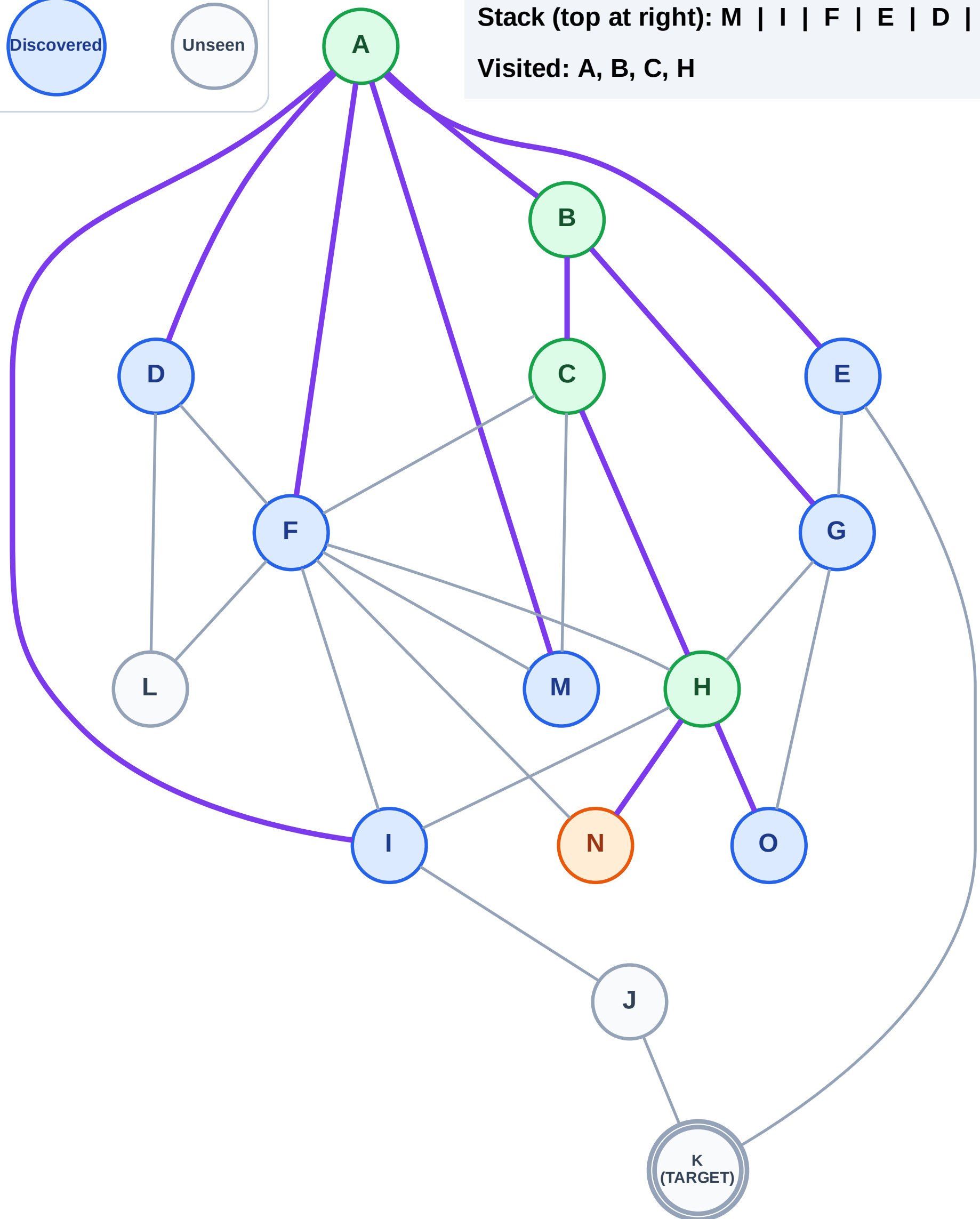
Step 10: Process node N

Legend



Stack (top at right): M | I | F | E | D | G | O

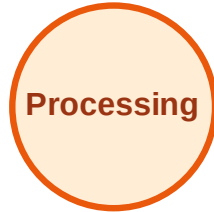
Visited: A, B, C, H



DFS Traversal

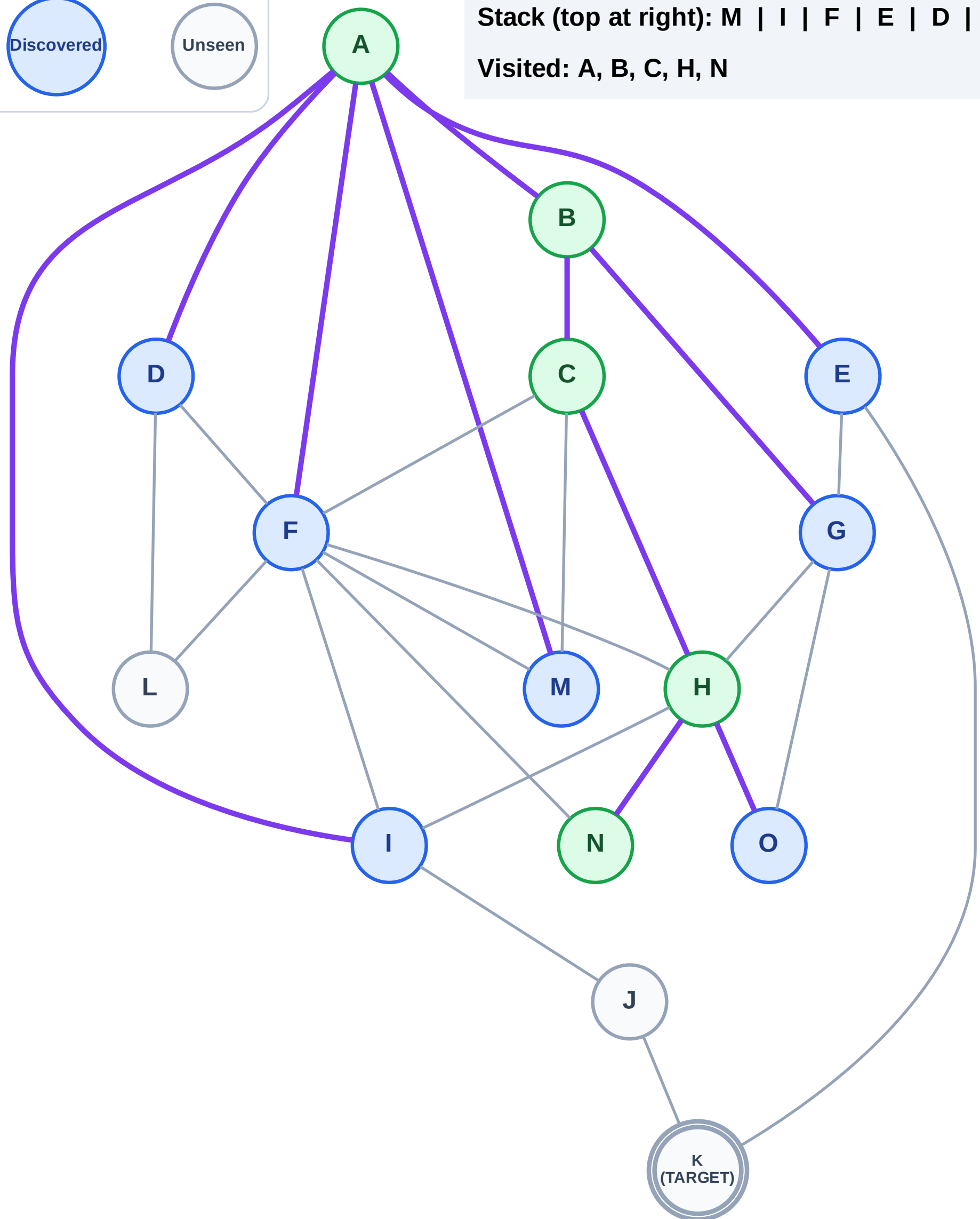
Step 11: No new neighbors from N

Legend



Stack (top at right): M | I | F | E | D | G | O

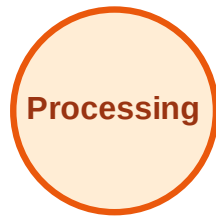
Visited: A, B, C, H, N



DFS Traversal

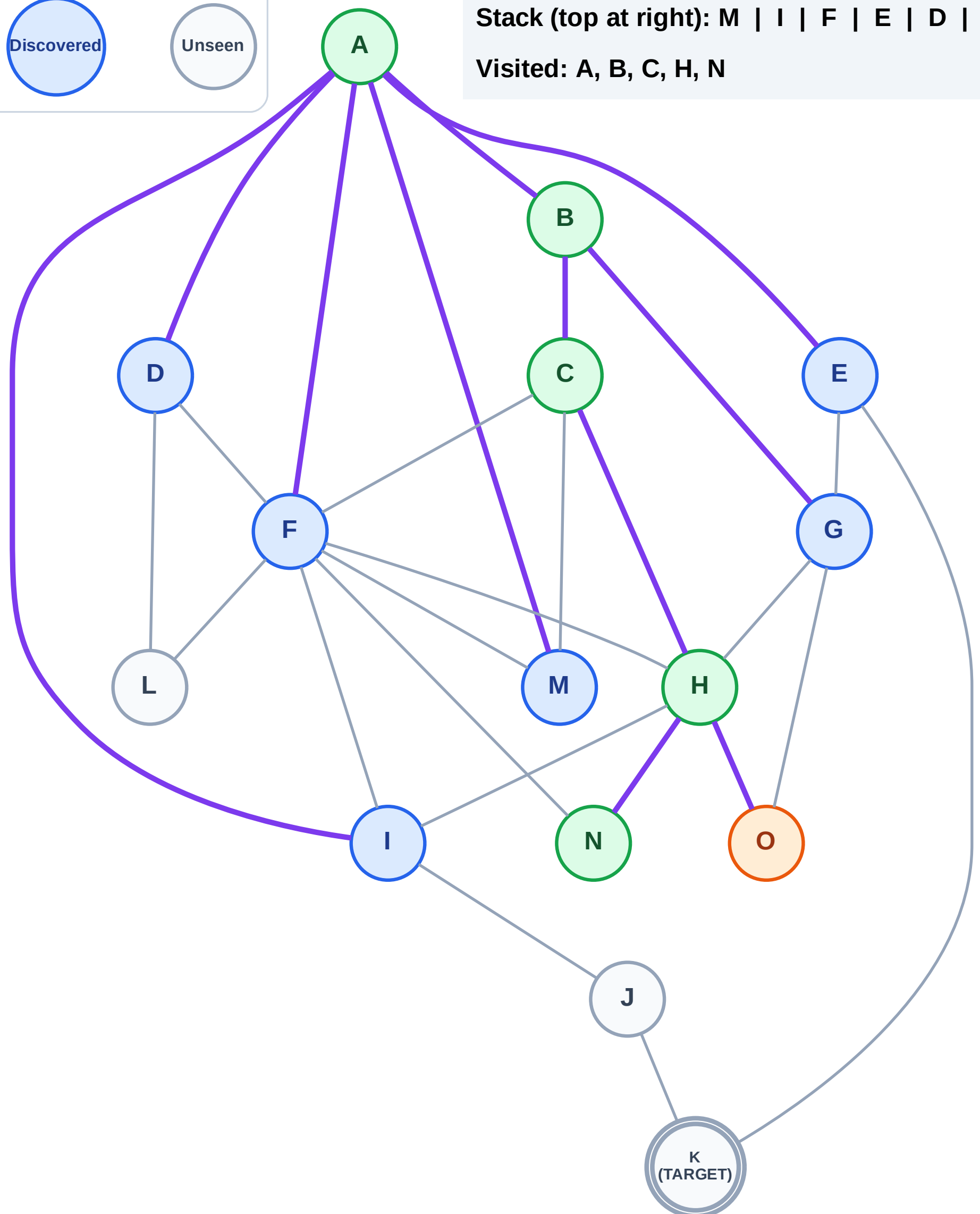
Step 12: Process node O

Legend



Stack (top at right): M | I | F | E | D | G

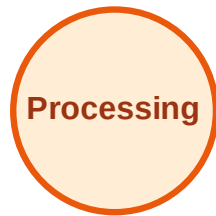
Visited: A, B, C, H, N



DFS Traversal

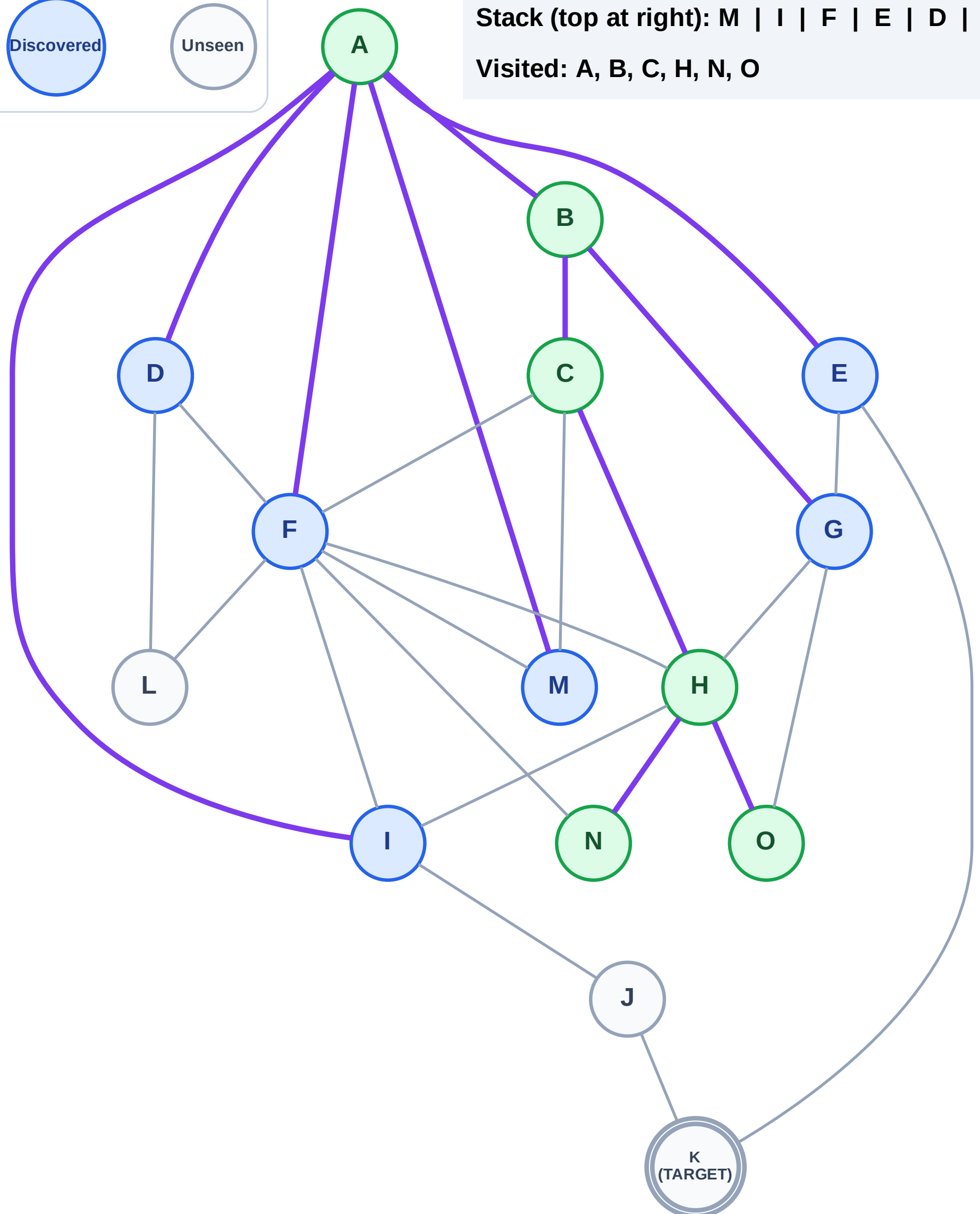
Step 13: No new neighbors from O

Legend



Stack (top at right): M | I | F | E | D | G

Visited: A, B, C, H, N, O



DFS Traversal

Step 14: Process node G

Legend

Complete

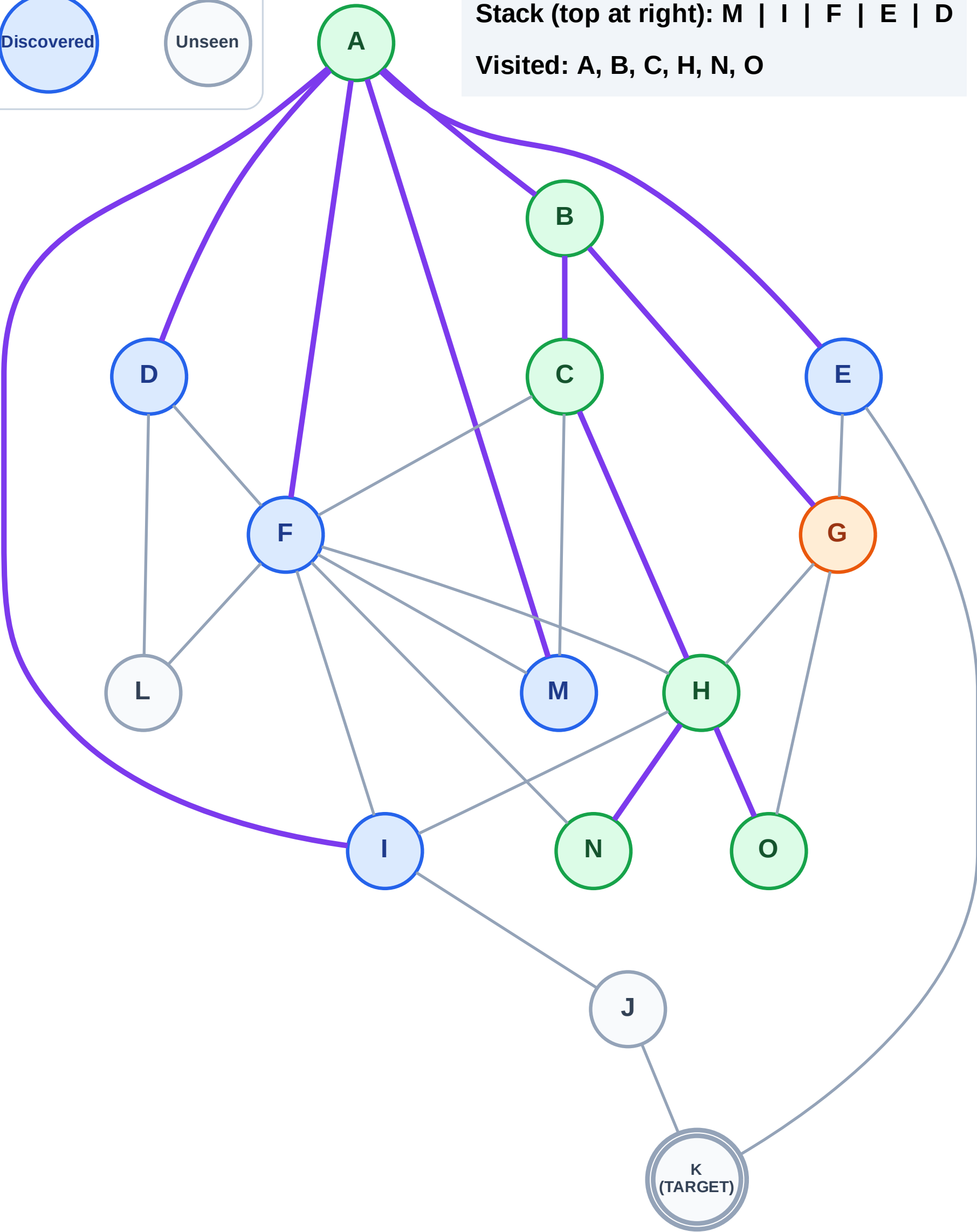
Processing

Discovered

Unseen

Stack (top at right): M | I | F | E | D

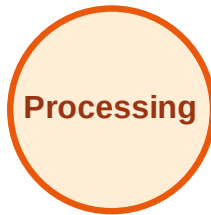
Visited: A, B, C, H, N, O



DFS Traversal

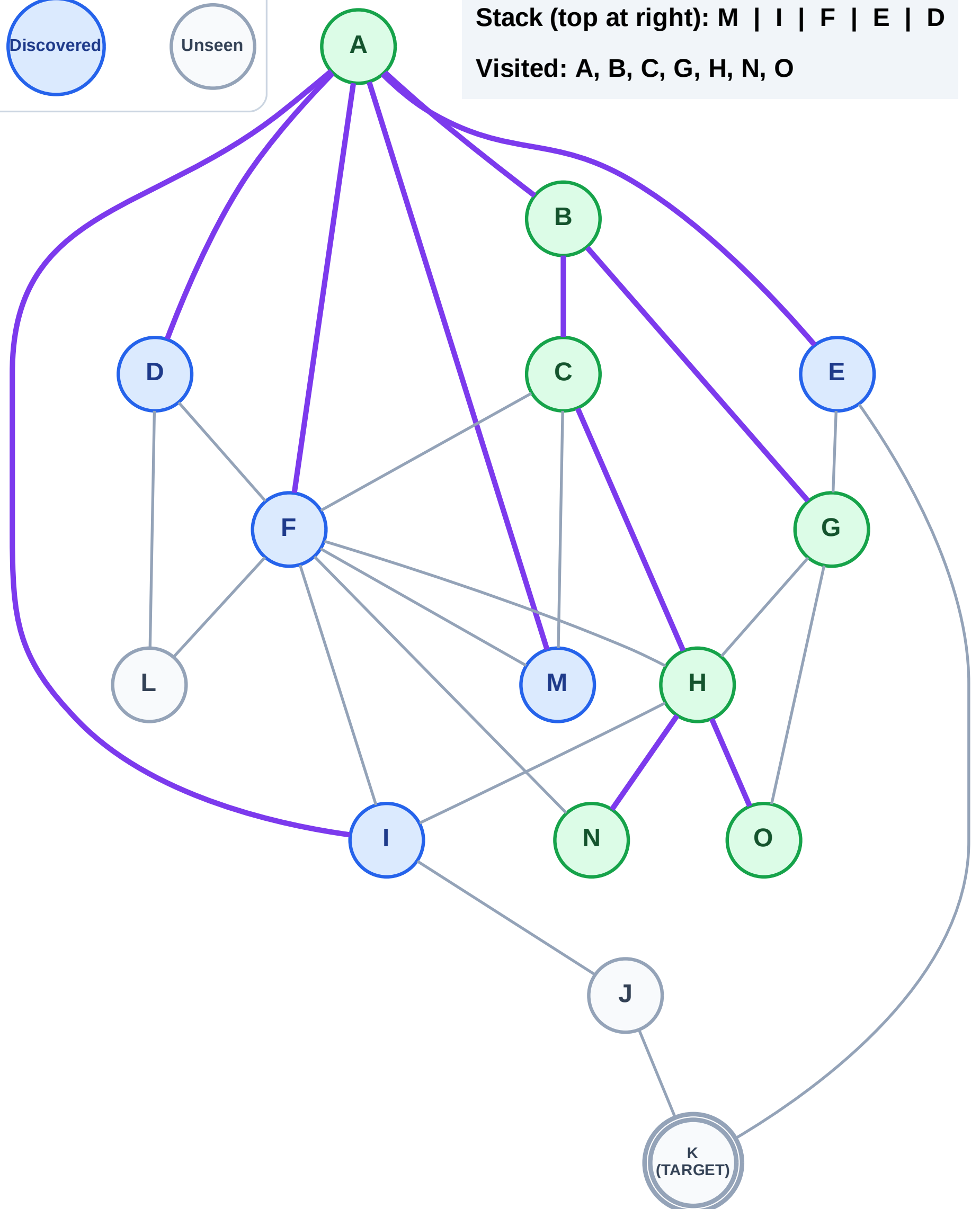
Step 15: No new neighbors from G

Legend

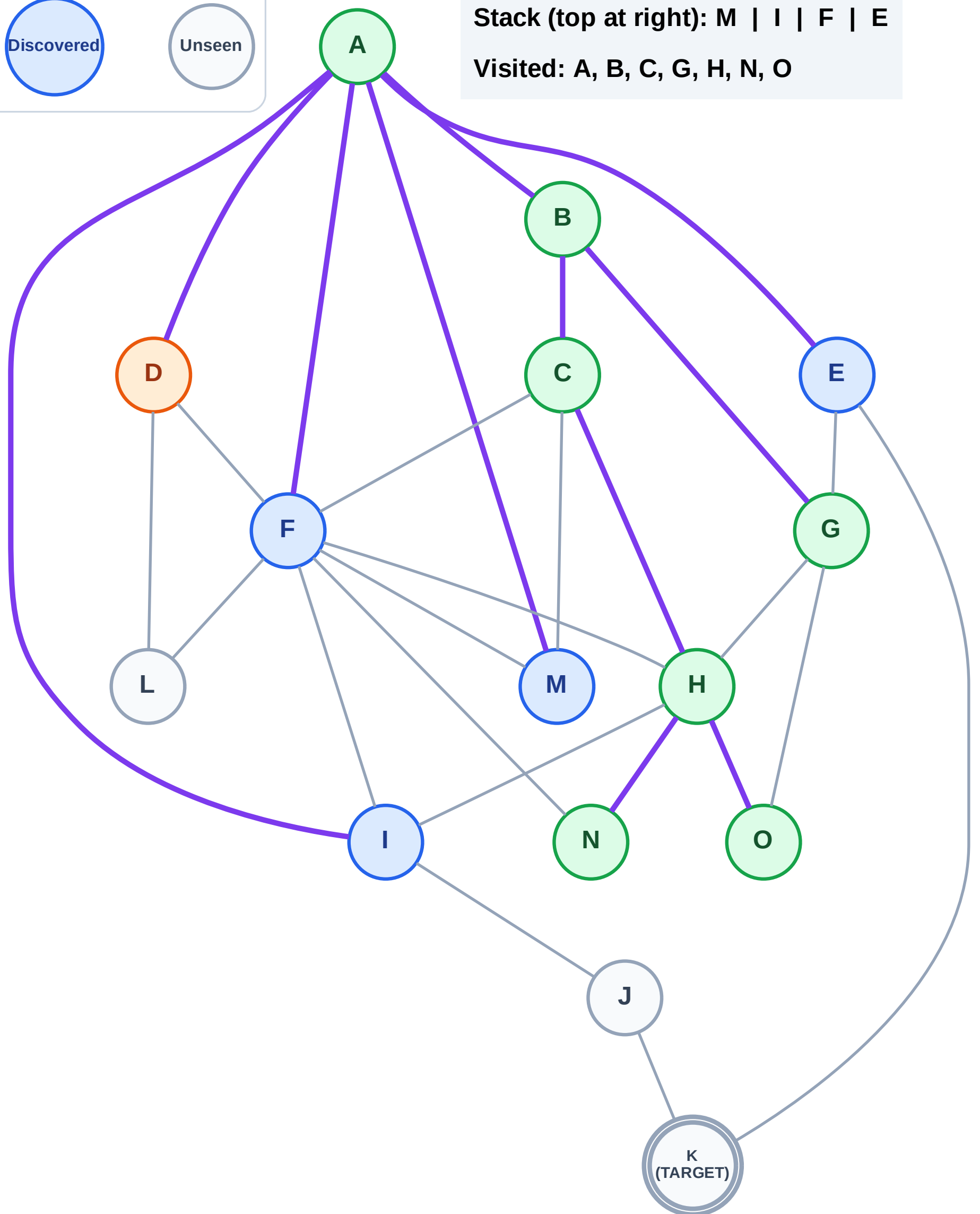
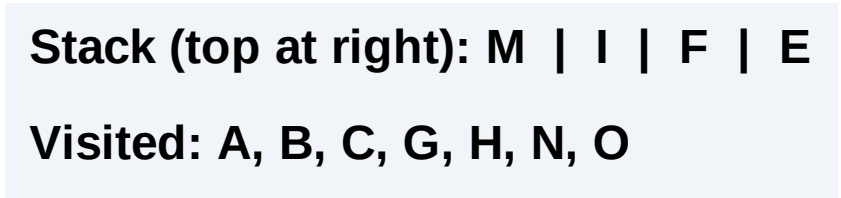


Stack (top at right): M | I | F | E | D

Visited: A, B, C, G, H, N, O



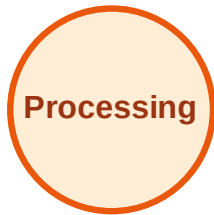
Step 16: Process node D



DFS Traversal

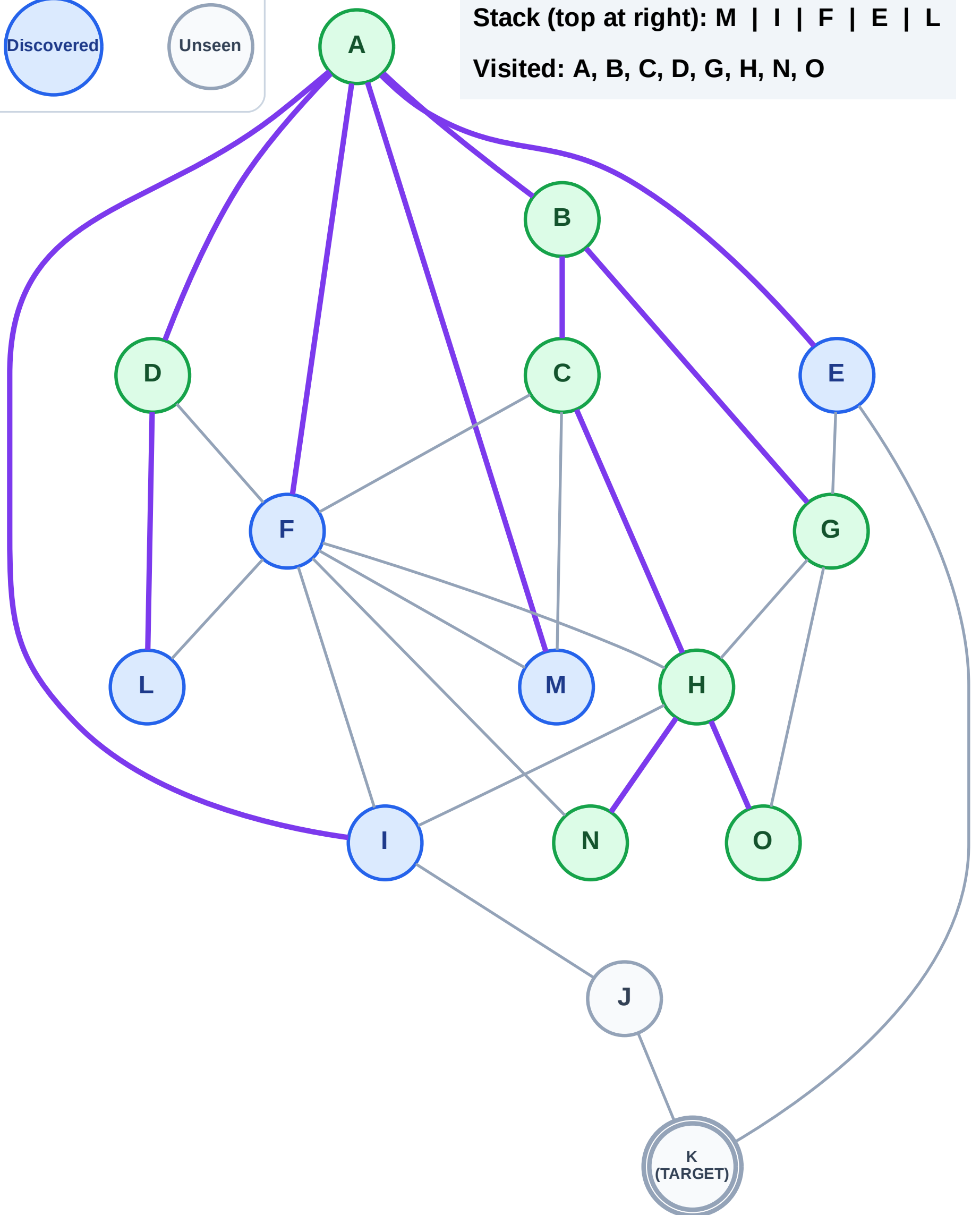
Step 17: Discover L from D

Legend



Stack (top at right): M | I | F | E | L

Visited: A, B, C, D, G, H, N, O



DFS Traversal

Step 18: Process node L

Legend

Complete

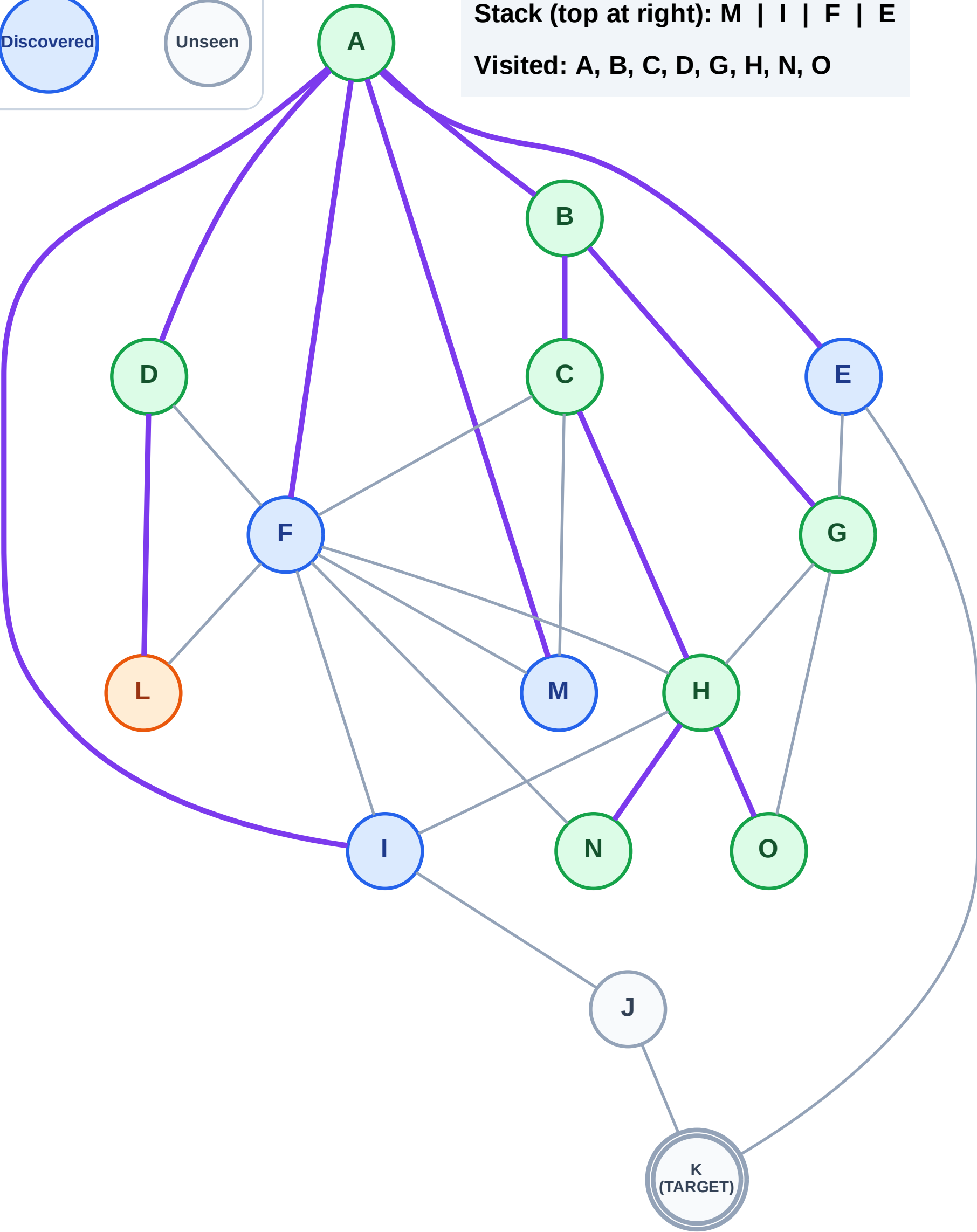
Processing

Discovered

Unseen

Stack (top at right): M | I | F | E

Visited: A, B, C, D, G, H, N, O



DFS Traversal

Step 19: No new neighbors from L

Legend

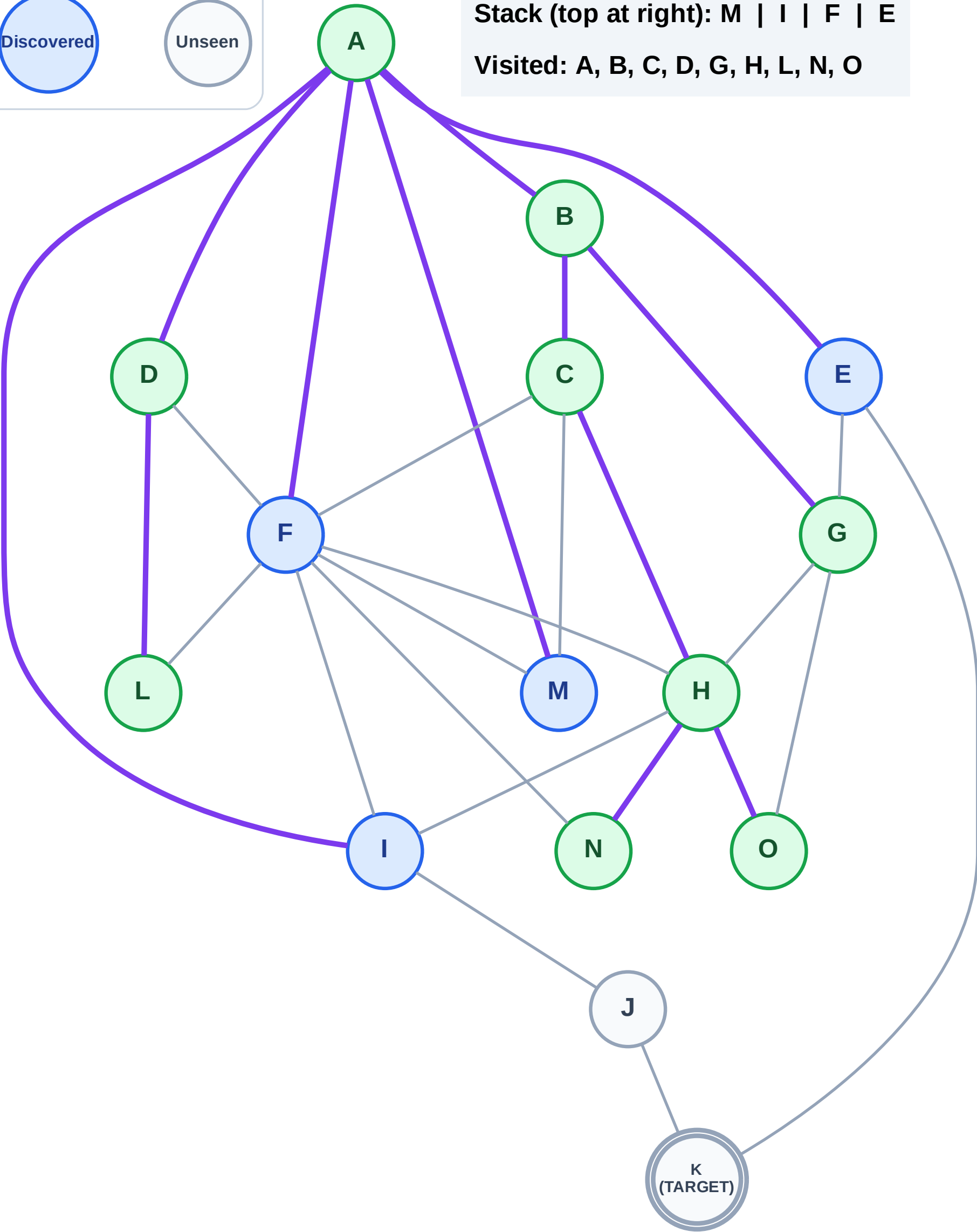
Complete

Processing

Discovered

Unseen

Stack (top at right): M | I | F | E
Visited: A, B, C, D, G, H, L, N, O



DFS Traversal

Step 20: Process node E

Legend

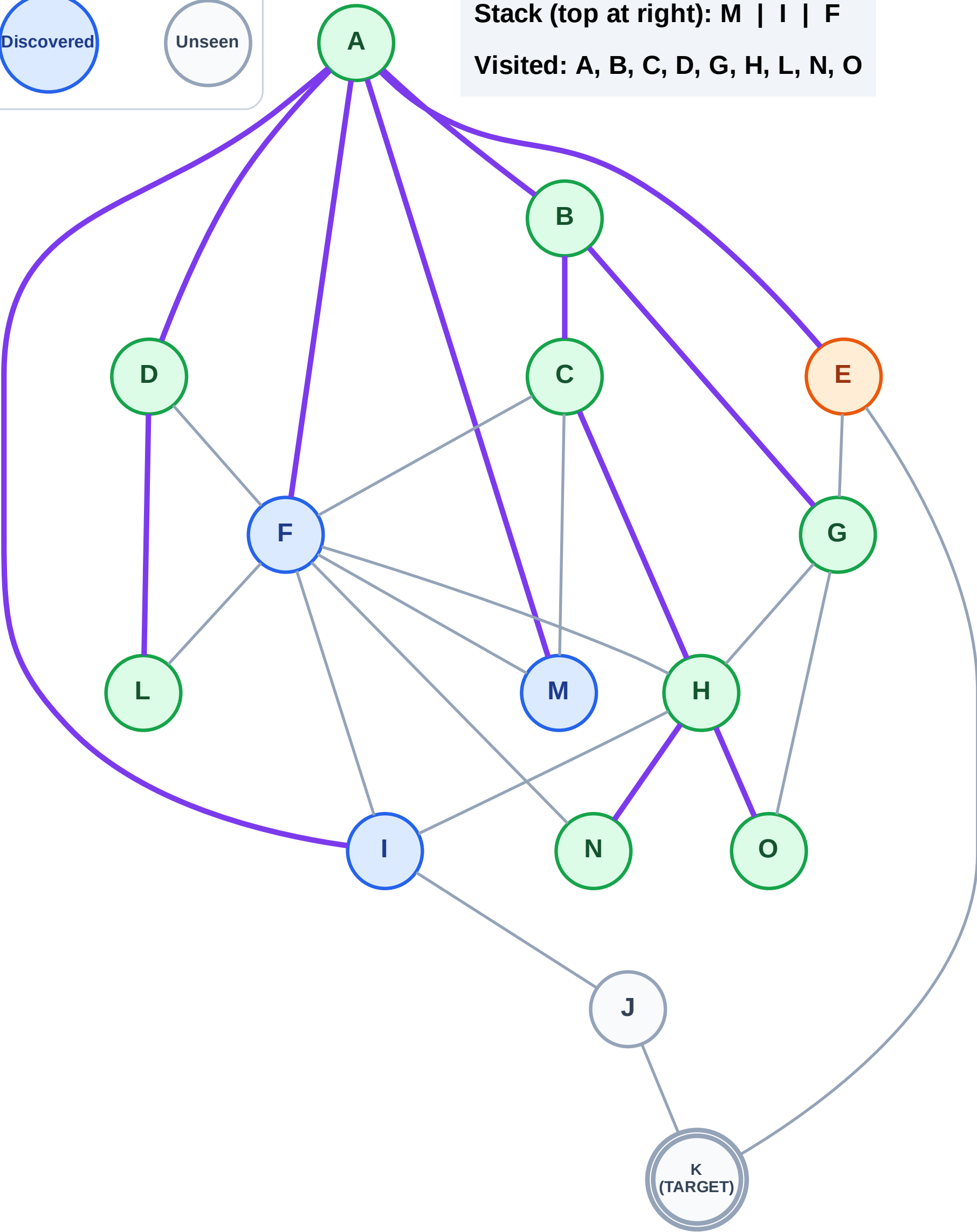
Complete

Processing

Discovered

Unseen

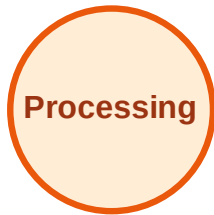
Stack (top at right): M | I | F
Visited: A, B, C, D, G, H, L, N, O



DFS Traversal

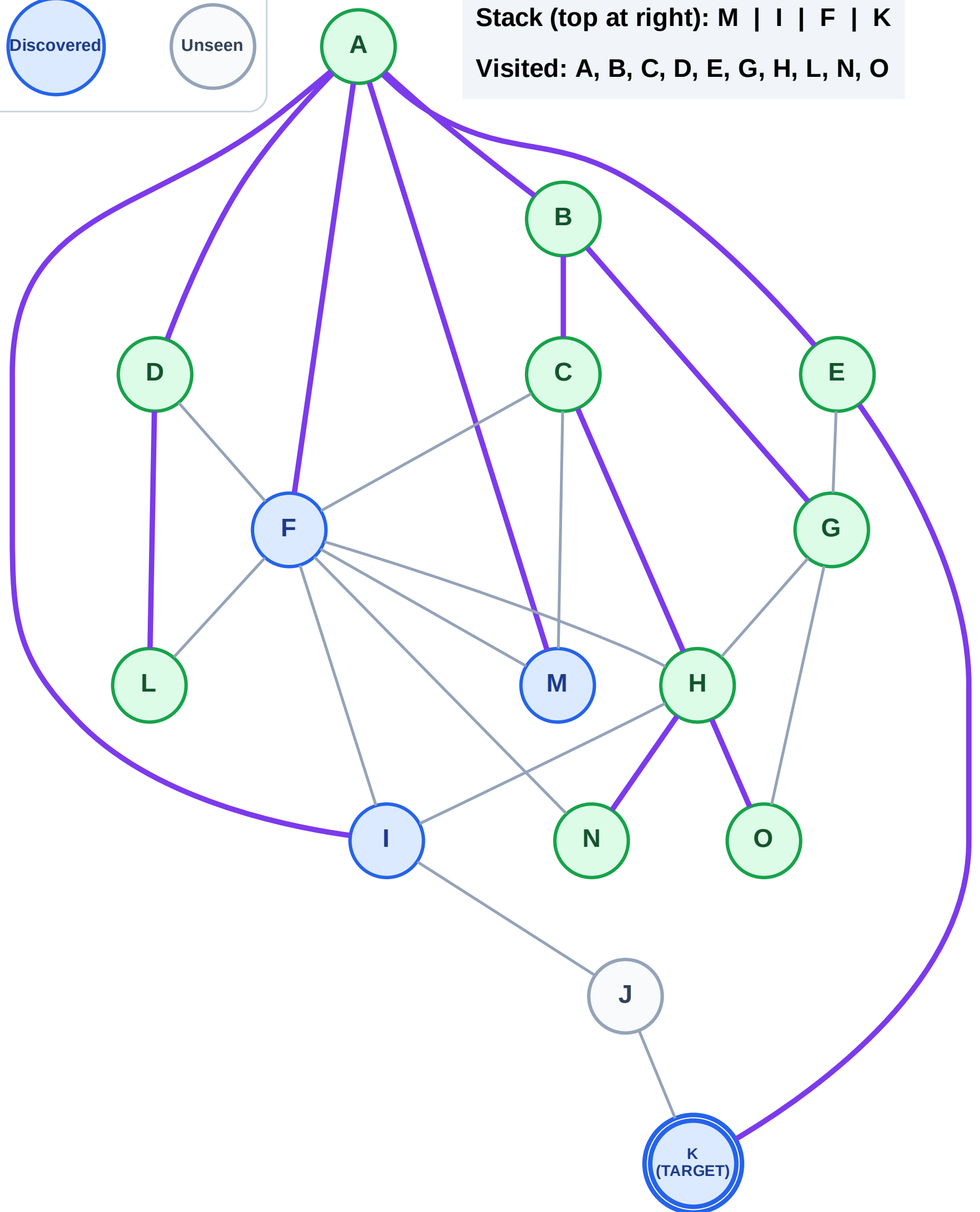
Step 21: Discover K from E

Legend



Stack (top at right): M | I | F | K

Visited: A, B, C, D, E, G, H, L, N, O



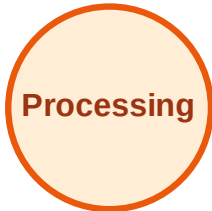
DFS Traversal

Step 22: Process node K

Legend



Complete



Processing



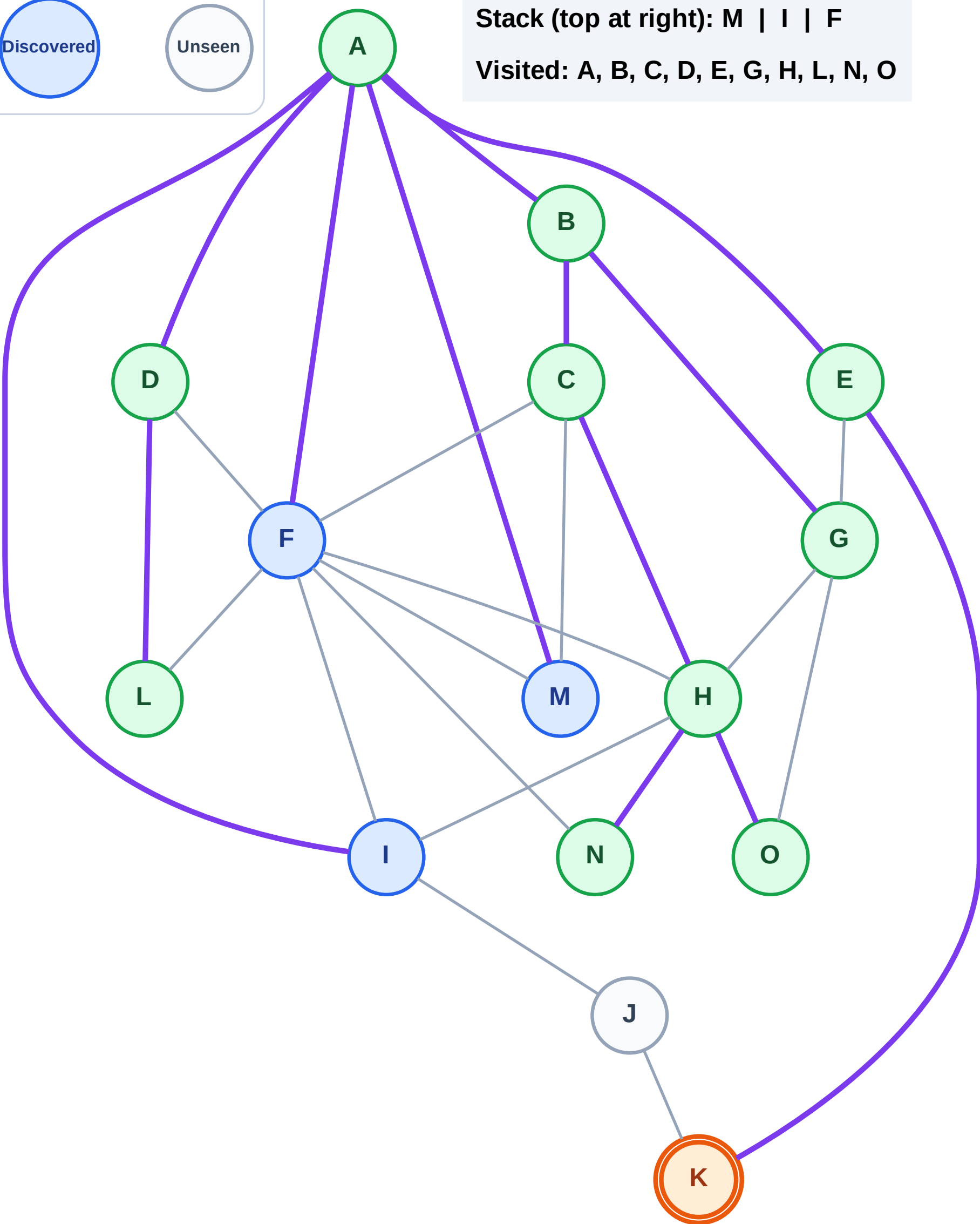
Discovered



Unseen

Stack (top at right): M | I | F

Visited: A, B, C, D, E, G, H, L, N, O



DFS Traversal

Step 23: TARGET NODE K REACHED! Path (2 hops): A → E → K

Legend

Complete

Processing

Discovered

Unseen

Stack (top at right): M | I | F

Visited: A, B, C, D, E, G, H, K, L, N, O

